



Transaction Security Services version upgrade guide BASE24-eps[®]

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About this document

The BASE24-eps[®] Transaction Security Services (TSS) process provides all hardware security module functions for BASE24 processes. The BASE24-eps TSS process is built using ACI's Enterprise Services object-oriented architecture. Files used by the BASE24-eps TSS process are maintained through a separate user interface. The **BASE24-eps Transaction Security Services version upgrade guide** describes how to upgrade existing BASE24-eps TSS systems to a newer version.

The upgrade requirements in this guide are organized by version. You must take into account all requirements between your existing BASE24-eps TSS version and the target version you are upgrading to. For example, if your current BASE24-eps TSS system is on version 02.4 and you want to upgrade to version 06.4, you must first upgrade the database from 02.4 to 03.1, then from 03.1 to 03.2, and so on.

On the other hand, if the current BASE24-eps TSS system is on version 06.2 and you want to upgrade to version 07.4, you need only perform the steps for upgrading from version 06.2 to version 06.4 and then to version 07.4.

If you are upgrading from a previous version to another previous version, for example, from version 03.2 to version 03.3, you need only perform the steps for upgrading from version 03.2 to version 03.3, and so on.

It is not necessary to perform a complete install of each intervening version of BASE24-eps TSS software when performing a multi-version upgrade. For example, when upgrading from version 03.3 to version 07.4, the database must be upgraded from version 03.3 to 04.1 to 04.4 to 05.2 to 05.4 to 06.4 to 07.4 (using the procedures outlined in the sections 6–8 of this document), but it is not necessary to perform a full install of the 04.1, 04.4, 05.2, 05.4, 06.2 and 06.4 versions of software.

Audience

This manual is intended for system administrators and operational staff who are responsible for the installation and configuration of the BASE24 transaction security environment.

Prerequisites

This manual should be used in conjunction with the **BASE24 Transaction security manual**, the **BASE24 Transaction Security Services processing guide**, and the context-sensitive online help provided with the ACI desktop user interface. The **BASE24 Transaction security manual** provides detailed information on BASE24 processing for transaction security functions including configuration, processing steps, and internal message content. The **BASE24 Transaction Security Services processing guide** provides detailed information on how the Transaction Security Services (TSS) process is integrated into a BASE24 system.

Also important is an understanding of the basic functionality of the hardware security modules to be integrated with BASE24 and BASE24-eps TSS.

Additional documentation

The following BASE24 publications can be referenced for additional information:

- The ***BASE24 Transaction security manual*** provides detailed information on configuration, processing steps, and internal message content as well as on using triple DES in a translate mode only in certain Interchange Interface processes without using the Transaction Security Services process.
- The ***BASE24-eps and TSS (v11.1) HP NonStop install guide*** provides detailed information on the steps required to install a new version 11.1 TSS system.
- The ***BASE24 Release 6.0 implementation guide*** and the ACI desktop user interface and Transaction Security Services Installation section of the ***BASE24 6.0 installation guide*** provide detailed procedures on installing the Transaction Security Services process and running the BASE24 database conversion programs.

What's new

The following table highlights the major changes that have been made in the most recent update to the **BASE24-eps TSS Version Upgrade guide**. The first column of the table lists the sections in which major changes have been made. The second column of the table describes the major changes for each section. These changes were made since the document was first published for BASE24-eps Release 1.0 version 11.1.

Section	Major changes
7	Replaces channel security database conversion information for earlier versions. This information was inadvertently left out when this guide was published for v11.1.
6 - 8	Sections 6, 7, and 8 were inadvertently changed from their original appendix names of A, B, and C when this guide was published for v11.1. Changed these sections back to their original appendix naming convention.
C	New fields added in v09.2 were renamed in v11.1. The DALCI inserts needed to be adjusted to instruct the use of the old names for an upgrade to v09.2 while the new names are needed for an upgrade to v11.1.

1: Getting started

It is essential that you read this entire document and carefully plan your migration before executing any implementation tasks.

Beginning with version 04.4, installation of the BASE24-eps Transaction Security Services (TSS) software (including all supporting components such as the UI Security Server) has been automated. It is important that all procedures outlined in the **BASE24-eps and TSS (v11.1) HP NonStop install guide** be completed as the first step in upgrading from a previous release of TSS to version 11.1.

Upon successful completion of the installation procedures, some additional manual steps are required to migrate an existing pre-11.1 version of the BASE24-eps TSS database to the 11.1 version.

Version 03.1 enhancements

Version 03.1 of the Transaction Security Services subsystem includes several enhancements and new features. These include the following:

- Dynamic key management for ATM and POS devices
- Encrypting PIN Pad (EPP) serial number validation for the Automated Key Distribution System (AKDS) add-on product
- Support for Atalla's Identikey PIN verification method
- Support for data encryption and decryption (as defined by the Canadian Financial Institution Layer (FIL) specifications)
- Support for the Eracom Mark II security module (ESM)
- Support for offline PIN change (as defined in the EMV specifications)
- New key activation

Dynamic key management for ATM and POS devices

Dynamic key management enables automated updating of transaction keys based on a number of factors. Thresholds can be established based on time, number of transactions, number of errors, number of consecutive MAC errors, or any combination of these factors.

Dynamic key management requires the use of hardware security modules to perform PIN encryption, PIN verification, and message authentication. This feature is not supported when these functions are performed in software.

Encrypting PIN Pad (EPP) serial number validation

The Automated Key Distribution System (AKDS) add-on product has been enhanced to optionally support the validation of the EPP serial number during the initial key exchange between the host (BASE24 or BASE24-eps TSS) and device. When enabled, the BASE24-eps TSS process performs additional validation to verify that the EPP serial number included in the initial key exchange request message is valid and known to the host system.

This enhancement also allows for the tracking of where a specified EPP is installed and when it was last validated (i.e., the last date and time that the initial key exchange was performed for the identified EPP).

In addition, specific log messages are issued to allow enhanced tracking of EPPs within a host system.

Identikey PIN verification method support

The Identikey PIN verification method support enhancement provides full support for the Identikey method of PIN verification. Functions supported include PIN verification and PVN generation using this method. Note that this verification method is only supported when using Atalla hardware security modules.

As a result of this enhancement, two new data sources are required in the BASE24-eps TSS database.

Data encryption and decryption support

The data encryption and decryption support enhancement provides support for data encryption, decryption, and translation functionality. Support is provided for the following commands:

- Encrypt Data (new command)
- Decrypt Data (new command)
- Translate Encrypted Data (new command)
- Encrypt Data and Generate Message Authentication Code (new command)
- Translate Data and Generate Message Authentication Code (new command)
- Generate Message Authentication Code using Initialization Vector (enhanced command)

These commands support multiple DES encryption methods, including the following:

- Cipher block chaining (CBC)
- Cipher feedback, eight bits (CFB-8)
- Cipher feedback, 64 bits (CFB-64)
- Output feedback, 64 bits (OFB-64)
- Electronic Code Book (ECB)

Note that not all DES encryption methods are supported by all commands. Specifically, encrypt/translate with MAC generation currently supports only CBC and ECB methods.

Where CBC method is used, the optional use of static Initialization Vectors is supported. Dynamic Initialization Vectors are not supported at this time.

The keys used for data encryption or decryption functions are located on the Channel Security and the Interchange Security data sources. As with other transaction keys, full support for dynamic key exchange is supported for this new key type.

As a result of this enhancement, both the Channel Security and Interchange Security data sources must be converted as part of the migration from version 02.4 to 03.1.

Eracom Mark II security module support

The Eracom Mark II security module support enhancement allows use of the Eracom Mark II hardware security module.

EMV offline PIN change support

The EMV offline PIN change support enhancement allows a cardholder to change offline a PIN held on an EMV card. It also supports offline PIN blocking.

In addition, EMV scheme retrieval from the EMV Security data source is also included in this enhancement.

EMV offline PIN change is supported when using the Atalla (standard and AKB formats) and Thales-e hardware security modules.

New key activation support

When a BASE24 interface generates a new key using BASE24-eps TSS, the new key does not become the current key until an acknowledgement is received from the host, co-network, or switch. When BASE24-eps TSS generates a new key, it updates the Interchange Security (ISEC) record on disk, but optionally does not change the current index value until the New Key Activate request is received. The New Key Activate request is a new BASE24-eps TSS command that is sent by the host, co-network, or switch when an acknowledgment has been received that the key has been successfully transmitted.

A new BASE24-eps TSS message is defined for the New Key Activate command (command 18). A new field is defined for the existing BASE24-eps TSS Key Generate Command (command 04) indicating when the current index value should be updated.

Version 03.2 enhancements

Version 03.2 of BASE24-eps TSS includes several enhancements and new features. These include the following:

- Dutch Security Enhancements
- iCVV Validation
- CVV and CVV2 Generation

Dutch security enhancements

The Dutch-specific HSM functions are used to support common security features for the Netherlands market. These features cover the following areas:

- Key Management for the BIM protocol used by the Interpay national ATM/POS switch (NBGC Interface)
- MAC generation and verification
- PIN verification using the NCR method

The Dutch security scheme for interface dynamic key management is based on a three-level hierarchy:

- Master key encryption key is a static key and is used to encrypt transport keys (transport keys are also key encryption keys)
- Transport keys are dynamic keys and are used to encrypt session (working) keys
- Session keys are used for the online transactions for PIN encryption and MAC generate/verify function

iCVV validation

The CVV check for chip cards uses the standard CVV processing as for magnetic stripe cards with the exception that the Service Restriction Code value is set to 999.

CVV and CVV2 generation

The DES algorithm used by the hardware security module (HSM) is the same for the generation of the CVV and CVV2 values. The only difference is in the data provided to the HSM; the CVV2 generation uses a value of 000 for the service restriction code, while the CVV generation uses the actual service restriction code for the card.

Version 03.3 enhancements

Version 03.3 of BASE24-eps TSS includes several enhancements and new features. These include the following:

- Enhanced Tracing of HSM Messages
- CAVV Validation
- Interac Key Management
- CONFCSV Changes

Enhanced tracing of HSM messages

To supplement the testing and problem determination processing for BASE24-eps TSS, the enhanced tracing of HSM messages enhancement enables the external message tracing for specific devices. The Message Delivery component has been enhanced to process device names from the DELIVER TRACE and ALTER TRACE commands.

CAVV validation

Cardholder Authentication Verification Value (CAVV) support is provided with this version.

Interac key management

The Interac key management enhancement provides support for requirements specified by Interac for key management processing. This includes the ability for the client application (typically BASE24) to specify the key slot to be used in key exchange messages (key generate and key store requests) as well as specify the key to be used for certain security operations.

CONFCSV changes

As a result of testing on multiple platforms, it was determined that the data source short names contained in the CONFCSV must be limited to 16 characters or less. To satisfy this restriction, all short names longer than 16 characters were modified to be 16 characters or less. This affects the following BASE24-eps TSS data sources:

Original short name	New short name
AUDIT_STORE_AND_FORWARD_CONFIG	AUDIT_SAF_CONFIG
CARD_VERIFICATION	CARD_VRFCTN
CHANNEL_PUBLIC_KEY_SECURITY	CHAN_PKEY_SEC
DIEBOLD_NUMBER_TABLE	DIEBOLD_NUM_TBL
EPP_SERIAL_NUMBER	EPP_SERIAL_NUM
HOST_PUBLIC_KEY_SECURITY_PRI	HOST_PKEY_PRI
HOST_PUBLIC_KEY_SECURITY_SCND	HOST_PKEY_SCND
INTERCHANGE_SECURITY	ICHG_SECURITY
NCR_PIN_VERIFICATON	NCR_PIN_VRFY
SECURITY_DEVICE_CONFIG	SEC_DEV_CONFIG

Changes were made to the Authentication, Channel Security, Interface Security and Security Device Configuration UIs to support a new HSM device type, Eracom AMB. The functionality provided by Eracom AMB differs from the current Eracom Mark II HSM and also has some unique encrypted key display and key entry requirements.

Version 04.1 enhancements

Version 04.1 of BASE24-eps TSS includes several enhancements and new features. These include the following:

- Derived unique key per transaction (DUKPT) support
- Check digit validation and generation utility
- Derived unique key per transaction (DUKPT) MAC support
- EMV derivation key index support
- Interac changes for BASE24 KEYF
- ISO Format 1 PIN block support

Derived Unique Key Per Transaction (DUKPT) support

This enhancement provides the ability to translate an inbound PIN block from DUKPT to DES for the Atalla and Thales HSMs.

Check digit validation and generation utility

The new check digit verification and generation utilities can be used by all customers who utilize the Atalla HSM modules and are migrating from variant format keys to the Atalla Key Block (AKB) format keys. These utilities provide the means to verify and generate check digits for all keys stored in the TSS database.

Derived Unique Key Per Transaction (DUKPT) MAC support

The purpose of this enhancement to BASE24-eps TSS is to include the security commands necessary to satisfy the BASE24 DUKPT MAC requirements.

Note that only Atalla (AKB) and Thales commands are supported at this time.

EMV Derivation Key Index (DKI) support

The EMV Security data source contains 24 derivation keys. Currently, the relationship between the derivation key index (DKI) and the set of 24 keys is implied. For example, if the DKI is 1, the first derivation key in the data source is used, if the DKI is 10, the tenth derivation key in the data source is used, etc. This assumes that the range of DKI values is 1–24, inclusive. However, customers have defined DKI values in the range of 0–255, inclusive. This enhancement implicitly adds the DKI field(s) to the EMV data source.

Interac changes for BASE24 KEYF

The Interac changes for BASE24 KEYF enhancement adds support for a new tab on the BASE24 KEYF UI. The new tab is named “Interac” and is displayed only if the Interac package is present.

The initial implementation is for the IMNI interface supported by the Canadian office. All Canadian customers that want to migrate to BASE24 Release 6.0 require this enhancement.

ISO Format 1 PIN block support

The ISO Format 1 PIN block support enhancement added the ability for Thales HSMs to support the ISO Format 1 PIN block.

Version 04.4 enhancements

Version 04.4 of BASE24-eps TSS includes several enhancements and new features. These include the following:

- AMEX Card Security Code (CSC) validation support
- Expiration date in card verify message
- Encryption type in key generate response
- Encrypted decimalization table support
- Disable public/private key generation (AKDS only)
- EMV2000 support
- Eracom AMB hardware security module support
- MFK/LMK conversion utility for public keys (AKDS only)

In addition to the functional enhancements listed above, the following delivery, implementation and operational changes have been included with version 04.4:

- Delivery of source code documentation
- Delivery of BASE24-eps TSS in Linkable Object Form
- Support for Format 2 files (HP NonStop only)
- NSJ4 support
- Enhanced support of versioned tables (SIS change)

AMEX Card Security Code (CSC) validation support

American Express (AMEX) has issued card verification guidelines for card issuers to follow with regards to the process of encoding cards with Card Security Codes (CSC).

American Express offers three types of CSC. The three types are as follows:

- Three-digit CSC, located on the signature panel
- Four-digit CSC, located on the front of the card
- Five-digit CSC, located on the magnetic stripe

BASE24-eps TSS has been enhanced to support AMEX card verification. The AMEX card verification algorithm differs from the Visa and MasterCard algorithms. While the Visa and

MasterCard algorithms utilize the same Atalla and Thales HSM commands, the AMEX card verification algorithm requires different Atalla and Thales HSM commands.

The HSM commands can contain any combination of the three types of CSC. Therefore, the BASE24-eps TSS card verify request message must support all three CSC values (i.e., three-digit, four-digit and five-digit). Also, the BASE24-eps TSS Card Verify response message must support individual response values for each of the three CSC types (e.g., it is possible for one CSC type verification to pass, while another CSC type verification might fail).

BASE24-eps TSS has been enhanced to support the Verify Amex CSC command and the verification keys associated with CSC card verification when using the following HSM models:

- Atalla HSMs using Variant Key Scheme. Command 35A (AMEX CSC Verification) requires one double-length key.
- Atalla HSMs using AKB Key Scheme. Command 35A (AMEX CSC Verification) requires an AKB formatted key encrypted under the MFK. This field contains a 74-byte value, or a key table index. Only CSC keys having the following headers are supported in this field: 1mXNE000, 1mXNN000, 1mXVE000, and 1mXVN000.
- Thales HSMs. Command RY (AMEX CSC Verification) requires one double-length key.

The MFK/LMK conversion utility has been enhanced to support the CSC verification keys.

The Authentication Profile UI Card tab was modified to add the UI support for the American Express card verification scheme (CSC).

A new field, Verification Algorithm, was added to the Card tab so that the user can select which type of keys are to be entered (Visa/MasterCard or AMEX) and the UI formats of the key display based on the verification scheme selected. Visa/MasterCard Verification Algorithm is the default selection.

Special formatting is only required for Atalla non-AKB and Thales HSMs so the new Verification Algorithm value is visible and utilized for only these types of security devices. Existing formatting for AKB format keys required no modification.

Expiration date in card verify message

Currently the BASE24-eps TSS card verify message contains a date field. For CVV/CVD messages, this field contains the CVD date. However, this date can be in YYMM or MMY format, so it cannot accurately be used to read the CARD_VERIFICATION_OLTP data source. This issue was recognized and a product solution made whereby positions 4–8 of the KEYA Group field in the message are treated as the expiration date. However, certain customers that want to use the entire 16 bytes of the KEYA Group field are unable to do so. This enhancement resolves this issue by adding a new Expiration Date field.

Note: For backwards compatibility, positions 4–8 of the KEYA Group field in the message must be used as the expiration date if the Expiration Date field is not present, or is spaces. The new Expiration Date parameter is not optional for AMEX CSC transactions.

The Current Date field in the BASE24-eps TSS card verify message contains the expiration date for CAVV messages. This field continues to be used to carry the expiration date for CAVV messages.

Encryption type in key generate response

The BASE24-eps IFX Device Handler process must return a valid encryption type value in a message to the device that contains a new key. The encryption type value specifies the encryption mode of how the new key is used.

Encrypted decimalization table support

The encrypted decimalization table support enhancement supports encrypted decimalization tables.

For Atalla, the clear decimalization table can be loaded into the HSM using a console. The HSM encrypts the decimalization table and stores it locally. The HSM returns the index to the memory location and that value is stored in the decimalization table field in the database.

For Thales, the HSM encrypts the clear decimalization table. The HSM returns the encrypted decimalization table and that value is stored in the database. The value of the encrypted decimalization table consists of 16 hex characters. For Thales, a mix of clear and encrypted decimalization tables is not supported.

Support was added for encrypted decimalization tables on the Authentication UI IBMDES tab. The current edit checks for the decimalization table were modified for Atalla and Thales HSM types to support the new encrypted decimalization tables. There were no changes to the decimalization table for Eracom.

Additionally, a new key type entry, Decimalization Table (LMK pairs 18/19), for Thales HSM type support was added to the Master File Key (MFK) Conversion UI.

Disable public/private key generation (AKDS only)

The Disable KEYGEN UI is accessible through the ACI desktop user interface grouped with the AKDS UIs under the Transaction Security menu option.

Within the Disable KEYGEN UI, the host name and device type can be selected. The UI then displays the current KEYGEN status—ENABLED or DISABLED. The status can then be toggled between ENABLED and DISABLED and the result saved by clicking on the Save icon. This saves the KEYGEN status of the given device in the Host Public Key data source.

If a KEYGEN command is attempted when the security device is disabled, the KEYGEN is not allowed and an event message is generated:

```
04-11-12;09:37:34.794 \ARGUS.$W4Y41      ACI.1502.1000      120
      Mgr PPD: \ARGUS.$W4YN  Generator: P4Y^XML01
TSEC: KEYGEN not allowed - disabled through the UI. Host = DLK1
      , Implementation = 0.
```

EMV2000 support

This enhancement provides the ability to support the new Thales commands and the extended Atalla commands that are related to EMV 2000. The major difference between EMV 2000 HSM commands and the existing EMV 1996 commands is an additional parameter, Branch/Height. This parameter is required in the new or extended HSM

commands and is used by the HSM in its session key derivation algorithm. Therefore, a new data element, Branch/Height, is added to the EMV Security data source.

Below are the commands supported in this enhancement.

The new commands for Thales HSM:

- KY -> Generate secure message with integrity and optional confidentiality and PIN change (similar to current KU command)
- KW -> ARQC (or TC/ACC) verification and/or ARPC generation (similar to current KQ command)
- K0 -> Verify encrypted counters – M/Chip 4

These Thales EMV commands KY, KW, and K0 are not supported using Async protocol.

The extended commands for Atalla AKB HSM:

- 350 -> Verify EMV ARQC and generate ARPC
- 351 -> EMV PIN change
- 352 -> Generate EMV MAC

Three new BASE24-eps TSS messages are defined to support the EMV 2000 related commands:

- TSS message 17 -> EMV verify encrypted counters
- TSS message 22 -> EMV 2000 ARPC generation
- TSS message 23 -> EMV 2000 script MAC generation or offline PIN change

The Authentication Profile UI EMV tab was modified to add the UI support for EMV 2000.

A new field was added to the EMV tab with the following possible selection values. The default value is the first option.

- Branch factor 2; Tree Height 16
- Branch factor 4; Tree Height 8

Eracom AMB hardware security module support

The Eracom AMB hardware security module enhancement of the TSS adds support for a new HSM type, the Eracom AMB. The functionality provided by the Eracom AMB HSM differs from the functionality provided by the Eracom Mark II HSM. Therefore, a new TSEC derived class exists for the new Eracom AMB HSM type.

The following are the new Eracom AMB functions supported by TSS. The new functions are initiated via TSS calls from the BASE24 product SECUTILS.

Function	AMB command	TSS message	Usage / comments
TERMKEYGEN1 (Generate Terminal Session Keys and roll KEK1)	3000/3500	A0	POS Device Handler
TERMKEYGEN2 (Generate Terminal Session Keys using KEK2 and roll KEK1/KEK2)	3010/3510	A0	POS Device Handler
NODEKEYGEN – 6.3 (Generate interchange session keys)	3A00	A7	Interface
MKEYGEN (Generate ATM Master Key)	3B20	04	ATM Device Handler
RTMK (Key Translation - Receive) (Translate received interchange session keys)	4500	A2	Interface
MACGEN - 6.3 and 6.4	5500	08	POS Device Handler and Interface
MACVERIFY - 6.3 and 6.4	5600	09	POS Device Handler and Interface
PINVERIFY 2000 (Transaction from interchange; ANSI PIN Block format; DES (IBM3624) Pin Verify)	6500	14	Authorization
PINVERIFY 2000 – VISA 6.3 (Transaction from interchange; ANSI PIN Block format; Visa PVV Pin Verify)	6501	16	Authorization
PINVERIFY - 6.4 (Transaction from POS AS2805 6.4 terminal; ANSI PIN Block format; DES (IBM3624) Pin Verify)	6510	A6	Authorization

Function	AMB command	TSS message	Usage / comments
PINVERIFY – VISA 6.4 (Transaction from POS AS2805 6.4 terminal; ANSI PIN Block format; Visa PVV Pin Verify)	6511	A6	Authorization
PVVGen - using given PIN	65B4	12	Authorization
PINBLOCKTRANS - 6.3 to 6.3 (PIN Block translation – interchange to interchange)	6600	13	Interface
PINBLOCKTRANS - 6.4 to 6.3 (PIN Block translation – POS terminal to interchange)	6610	A3	Interface
KVC Request (Return Verification Code of a key)	7510	20	Interface and POS Device Handler under AS2805 6.2
CVVVERIFY	8520	01	Authorization
TERMPROOF - 6.4 (Proof of end-point processing – POS terminal)	E500	A1	POS Device Handler
NODEPROOF (Proof of end-point processing – request – interchange)	E520	A4	Interface
NODERESP (Proof of end-point processing – response - interchange)	E530	A5	Interface

Changes were made to the Authentication, Channel Security, Interface Security, and Security Device Configuration UIs to support a new HSM device type, Eracom AMB. The functionality provided by Eracom AMB differs from the current Eracom Mark II HSM and also has some unique encrypted key display and key entry requirements.

MFK/LMK conversion utility for public keys (AKDS only)

The MFK/LMK conversion utility for public keys enhancement adds support for the AKDS host public and secret keys or the channel public keys. Customers licensing AKDS that have audit requirements for changing the LMK/MFK require this enhancement.

The existing MFK/LMK key conversion utilities and UI windows have been modified to include the host public and secret keys and the channel public keys in the conversion from an old MFK/LMK to a new MFK/LMK.

Delivery of source code documentation

Source code is distributed to all customers for the business components that are licensed by the customer.

Source code is included on all distributions: initial install, upgrades, and updates/fixes. The source code must match the object code at the customer site.

Features include the following:

- The source code includes header and source files for the business components and the user interface components associated with these business components.
- The directory structure for the source code is the same as the product directories.
- The source code is distributed in zip files. An update/fix distribution should only include zip files that changed.
- The customer must not modify product classes. The customer can create new classes that extend product classes. The use of the source code is enforced by contract.

Delivery of BASE24-eps TSS in linkable object form

To support the planned customization strategy for BASE24-eps TSS, the individual business components are delivered in static library (.tlo) form that can be used by the HP Native Mode Linker (NLD) program to build the final executable.

Modifications to the BASE24-eps TSS Automated Installation program have been made to perform the following functions:

- Transfer of the individual static library objects from the installation CD or zip file to the HP NonStop in lieu of transferring the final executables.
- Execution of the NLD utility on the HP NonStop platform to build the final executables from the individual static library objects.

Support for Format 2 files (HP NonStop only)

The MetaMan application can generate output that is used to create individual tables or complete databases for the various supported databases. When BASE24-eps TSS is installed on the HP NonStop platform, the Enscribe database is used. The output that the MetaMan application generates can be used as input to the FUP utility to create database files of the various formats.

To create a Format 2 file, the MetaMan output must be changed slightly for FUP to create a Format 2 file. When a system is delivered, all of the database files are created using the standard Enscribe file type; however, using the instructions in this document, any database file can be created with type Format 2 so that it can exceed the 2 GB file size limit.

NSJ4 support

The major changes to support NSJ4 within the ACI desktop user interface pertain to the installation of the Java components on the HP NonStop platform. There are two major components that are used by the ACI desktop user interface:

- The ESWeb process under XPNET. This component is also known as the Satengine.
- The Java executable used by the USEC and VersionChecker Server processes.

The components that allow Java programs to access DAL data sources stored within Enscribe files (i.e., the JDAL JDBC Driver) were uplifted to be NSJ4 compliant. These JDAL components must be bound into the Java executable and the Satengine to facilitate access to the DAL data sources. For NSJ4, the changes include a different set of calls that handle TMF transactions than the previous releases of Java.

Because of these changes, modifications to the installation procedures are required for NSJ4 when supporting a JDAL.

Enhanced support of versioned tables

BASE24-eps TSS has been enhanced to support versioned tables within SIDAL. Unless otherwise stated, the information below refers only to versioned tables. The functionality of legacy tables is not changed with this enhancement.

Previously, elements could only be added to the end of a table, and only if a variable length element did not already exist. This enhancement eliminates, where possible, the need to recompile applications or convert existing tables when new columns are added to a table.

Note

This enhancement impacts the application performance for those tables that have more than one version. Significant overhead is required by the SIS level code when reading records for previous versions. For TSS Version 04.4, the Channel Security (CSECD) and EMV Security (EMVSD) data sources are at (data source) version 3.0.

When migrating to version 04.4 and later, it is recommended that a global update of the Channel Security data source be performed to update all rows to the current version and avoid any performance impacts.

Version 05.2 enhancements

Version 05.2 of BASE24-eps TSS includes the following enhancements and new features:

- CSEC memory enhancement
- CSEC Refresh Process Control command
- Additional security device configuration parameters
- HSM automatic reconnect

CSEC memory enhancement

This solution allows an unlimited number of Channel Security (CSEC) records to be supported within a BASE24-eps TSS system. Currently, due to memory and heap limitations, it is not possible for a single process to maintain all CSEC records in memory if the terminal population exceeds a maximum number of devices (on the HP NonStop, the current maximum per process is approximately 20,000 CSEC records).

Two new environment parameters, CSEC_MEMORY_USE (default Y) and MAX_CSEC_IN_MEM (default 20000) are used to determine whether the system should maintain a memory-based table of CSEC records and, if so, how many devices this table should support, or whether it should always read the record from disk.

CSEC Refresh Process Control command

Users are now able to warmboot a single CSEC record using the new CSEC REFRESH XXXX command from Process Control. This causes the system to reload the information for the Channel Security device specified by XXXX.

Additional security device configuration parameters

To make easier the support of both current and future combinations of security devices, protocols, and hardware platforms, the current field Device Subtype has been replaced by the following new fields on the Security Device Configuration window:

Hardware model

Valid values:

Atalla Ax000

Atalla Ax100

Thales RG7400

Thales RG71x0

Thales RG73x0

Thales HSM8000

Eracom Mark II

Eracom AMB

Protocol/interface

Valid values:

Async – NSK

Ethernet - Blocker NSK

TCP/IP – Boxcar NSK

TCP/IP – RAM NSK

TCP/IP – ICE-XS UNIX

TCP/IP – Adapter IBM

UDP/IP – URAM NSK

SDLC – HIP NSK

Key scheme

Valid values:

N/A

AKB – Atalla

Variant - Atalla

ANSI X9.17 – Thales

Variant – Thales

The following new fields have also been added to the Security Device Configuration window:

- Firmware Version

This field is set online during device initialization and is for display only. It shows the current firmware version of the security device.

- Max Buffer Length

This field is set online during device initialization and is for display only. It indicates the maximum length of a command message sent to the security device.

HSM automatic reconnect

The HSM automatic reconnect enhancement removes the need for TSS customers to issue an ENABLE command from the Process Control window when an HSM has been marked as out of service. The reconnection now is automatic. A connection is reattempted automatically at the interval of seconds specified by the new field, Reset Timer Limit, in the Security Device Configuration window, until the HSM is back in service.

Version 05.4 enhancements

AKDS Wincor support

Provide AKDS support for Wincor devices running in NDC and 912 emulation modes.

This enhancement includes the following new features in the TSS / AKDS product:

1. Support for a Wincor Signature device type.
2. Modifications to the AKDS UI work flow to accommodate the Wincor Signature AKDS implementation.
3. Support for a configurable Host name for the Host Public Key Security data sources.
4. Support for a new set of TSS AKDS messages (i.e., 52, 53, and 54).
5. Support for an EPP Serial Number with a maximum length of 16 bytes (current maximum is 8 bytes).
6. Support for TSEC functions for AKDS for BASE24-eps ATM device handlers (non-IFX).

AKDS Capacity report

This enhancement adds support for generating a report based on the number of devices configured in the Channel Public Key Security data source. A Process Control command can be issued manually to generate the report. The report is generated with the total number of devices and optionally with every Device ID listed. The AKDS Capacity Report is written to a write-only data source with the assign name AKDS_CAPACITY_REPORT.

Version 06.2 enhancements

APACS MACing support

The APACS POS terminal implements a transaction security scheme that provides keys that are unique to each transaction and that does not require long-term secret keys to be held in terminal equipment. Separate keys are used for Message Authentication and PIN Block Encryption. This scheme produces transaction keys which are updated for each transaction, and which are based on the key for the previous transaction, the current transaction contents, and the contents of the previous transaction. Thus transactions are chained providing automatic confirmation of previous transactions.

The BASE24-eps Transaction Security component has been modified to support the APACS Transaction Key scheme. Support will be provided for terminals driven by BASE24-pos and also terminals driven by BASE24-eps.

X9.19 MACing Support

The purpose of this enhancement is to support both ISO 9797 MAC algorithms for Thales and Eracom HSMs. Support is already in place to support both MACing algorithms for the Atalla HSMs.

TSEC currently supports three Thales MACing commands:

- MK – MAC Generate (ISO 9797 MAC Algorithm 1 - triple DES CBC)

- MM – MAC Verify (ISO 9797 MAC Algorithm 1 - triple DES CBC)

- MS – MAC Generate/MAC Verify (ISO 9797 MAC Algorithm 3 – pseudo triple DES CBC)

Support for command MS was originally added in support of the requirements for the Dutch region. For non-Dutch customers, there is no easy way to configure the ISO 9797 MAC Algorithm 3 (pseudo triple DES CBC).

This enhancement will allow the CSEC/ISEC MAC Type value to determine whether command MK/MM or MS is used.

Changes have also been made to Eracom Mark II and AMB modules. The MAC Generate and MAC Verify request formatting functions have been enhanced to support a MACing algorithm configuration based on the MAC Type parameter.

VISA DUKPT MACing support

Atalla command 386 (Generate DUKPT MAC) has been modified to support two additional MAC Type (algorithm) codes. The "VL" code uses the Visa DUKPT algorithm and returns the left half of the calculated MAC. The "VR" code uses the Visa DUKPT algorithm and returns the right half of the calculated MAC.

TSS Message 50 (DUKPT MAC Generate) will be modified to contain a new optional input parameter (MAC Option). The MAC Option parameter indicates whether the left half or right half of the calculated MAC should be returned by the HSM. According to the Scandinavian DUKPT MACing implementation, the left half of the MAC is returned for approved transactions and the right half of the MAC is returned for denied transactions.

Version 06.4 enhancements

Contactless chip and magnetic card (dCVV and CVC3) support

Support for Visa dCVV and MasterCard CVC3 (Dynamic Card Verification) is being added to BASE24-eps. BASE24 endpoints are being modified to capture dCVV/CVC3 data. TSS/TSEC has been modified to support new HSM commands that verify dCVV/CVC3 values.

A new TSS message has been defined to provide dCVV/CVC3 support for BASE24. The algorithms for Visa dCVV and MasterCard CVC3 differ as to what data is required. Therefore, the TSS message format will contain fields that are required for both dCVV and CVC3, fields that are only required for dCVV and fields that are only required for CVC3.

The following new Atalla commands will be supported:

- 357 (Visa dCVV Verify)

- 359 (MasterCard CVC3 Verify)

The following new Thales command will be supported:

- PM (Verify Dynamic CVV)

Thales command PM is used for both Visa dCVV and MasterCard CVC3 verification.

Contactless EMV chip card support will be added for Visa's VSDC program. The authentication uses the standard EMV 1996 ARQC Verification HSM commands.

Note: *Visa Smart Debit Credit (VSDC) Technical Guide to Visa's Applet for GlobalPlatform Cards*, section 9.4.1, states that Visa's VSDC applet implementation will only support PANs of 16 digits or less, so option B (EMV 4.1 Book 2 Option B Method) is not applicable to these implementations.

CVV1 and CVV2 combined verification support

Support for validation of both CVV1/CVC1 and CVV2/CVC2 when present in a single transaction is being added to BASE24-eps/TSS. The Transaction Security component and supporting components have been modified to validate CVV1/CVC1 and CVV2/CVC2 by making two calls to the HSM for a single transaction. This enhancement supports the validation of both types of cards: Admin cards and Consumer cards.

DEP (Banksys) host security module support

This project includes the following enhancements to BASE24-eps:

- Support for a new device type for the DEP HSM.
- Modifications to current data sources to store keys as fixed binary instead of fixed string.
- Support is provided for the DS3 message format. Support for the DS2 and DS4 message formats is not included.
- The following functionality is supported:
 - DES key management including Key Translate and Key Generate. Single, double, and triple length keys will be supported.
 - Card verification. Methods supported include CVV, CVC, CAVV and CSC.
 - Card verification generation. Methods supported are limited to CVV and CVC.
 - MAC verify and generate. Methods supported will be 1DES, 3DES and Pseudo 3DES (X9.19).
 - PIN processing support. Verify or generate using either IBM DES or VISA PVV methods. PIN Translate will also be supported.
 - EMV Support including ARQC Verify/ARPC Generate, Verify Encrypted Counters, Script MAC Generate, and Offline PIN Change. Both EMV'96 and EMV2000 will be supported.
 - RSA support including RSA Key Generate, Signature Verify, Signature Generate, HASH Generate, Import Public Key, and ATM Master Key Generate.
 - DUKPT support including PIN Translate (DUKPT to ANSI or PIN/Pad formats), DUKPT MAC Generate, and DUKPT MAC Verify.
- EMV Common Core Definition (CCD) support will not be included in this implementation.

EMV Common Core Definitions (CCD) support

This enhancement adds support for the modified Atalla and Thales EMV commands that include modifications for CCD.

A new value (CCD) will be defined for the EMV_SCHEME element in the EMV Security table.

Re-package TSEC

Product Management and the Channels have agreed that TSEC/TSS should be licensed in a manner consistent with BASE24 HSM support. Currently, BASE24 licenses HSM support separately by brand/model. For example, there is a separate license for Atalla support, Eracom Mark II support, Thales support, etc. BASE24-eps, however, currently bundles all support into a single licensable package.

This work effort will re-package TSEC/TSS into a "base" component and optional vendor-specific components. This will require changes in the management and packaging of TSS components in both the Development organization and the Software Management organization.

New customers will only receive the HSM code that they have licensed. Existing customers will continue to receive all HSM code until such time that their contracts are updated to reflect the new product code(s).

Regardless of the code received, installation procedures will be transparent.

HP NonStop object/executable size

On the HP NonStop platform, object sizes are significantly larger due to using the Version 3 compiler rather than the Version 2 compiler. HP documentation states the following:

"Cplusplus Version 3 library is more standards compliant, has better template support and tighter type checking so there are additional initialization functions and more globals. This extra text and data correspondingly lead to more symbols and thus produce a larger object file. This is normal."

Version 07.4 enhancements

Eracom Mark II EMV CCD support

This enhancement adds Common Core Definition (CCD) support for Eracom's Mark II security device. This is a continuation of the EMV Uplift initiatives.

Support for the following Eracom Mark II commands have been added for EMV CCD:

EE2018 – EMV_VERIFY_AC_GEN_ARPC

EE2007 – EMV_SCRIPT_CRYPT0

EE2016 - EMV_PIN_CHANGE_UNBLOCK

Eracom Mark II EMV uplift

This enhancement will extend Eracom Mark II security device support to include commands to support the EMV chip card scheme. EMV 2000 is supported. There is no support for EMV 1996 for the Eracom Mark II HSM.

Support for the following Eracom Mark II commands have been added for EMV 2000:

EE2018 – EMV_VERIFY_AC_GEN_ARPC

EF2013 – EMV_SCRIPT_CRYPTO_EMV2000

EE2017 – EMV_PIN_CHANGE_UNBLOCK_EMV_2000

Eracom Mark II keys on CSEC uplift

Currently, Eracom Mark II supports only HSM stored Master Keys, where the user enters a key index on the Channel Security (CSEC) UI, and the actual key is loaded in the HSM. This enhancement will allow customers to load the key's cryptogram on the CSEC, rather than loading it in the HSM.

Eracom proof of endpoint uplift

This enhancement adds support for the "proof of endpoint" authentication scheme specified in the AS2805 standard, which is supported by most Australian major banks.

Proof of Endpoint processing is optional in the AS2805 standard, however, most banks are requiring this method of endpoint authentication.

This will require two additional Eracom Mark II commands not currently supported:

EE3033 - NODEPROOF

EE3034 – NODERESP

Eracom Mark II data encryption/decryption product uplift

This uplift provides data encryption and decryption support. This will require two additional Eracom Mark II commands not currently supported:

EE0800 – ENCIPHER_2

EE0801 – DECIPHER_2

Both of these commands support Electronic Code Book (ECB) and Cipher Block Chaining (CBC) modes of operation.

Large message MACing

This will allow Message Authentication (MACing) for large messages that exceed the single message limits of a security device. This will be of value to customers handling ATM statement printing, large EMV issuer scripts, or other large messages that must implement MACing for compliance with regional security mandates. The enhancement will chain together multiple MACing commands.

The following lists some restrictions for large message MACing:

EMV Script Data TDE — Although changes will be made to support MACing of larger EMV Script Data, the existing EMV Script Data TDE has a limitation of 256 bytes on the length of the data. Additional modifications to the EMV Script TDE will be required to fully support MACing of larger EMV scripts.

SAMA TSS public key uplift

This enhancement will provide a facility to perform mutual authentication using digital signatures. The initial implementation will be for interfaces that support digital signature authentication during logon/logoff. The basic facilities will also be available for future signature-based authentication functions.

It is planned that Member Banks will use the standard AKDS product to fulfill the SAMA recommendations regarding ATM key management and will, therefore, eventually license the AKDS product. However, the SPAN2 Interface solution should be available to Member Banks regardless of whether they have licensed AKDS. Therefore, this SAMA solution should not be dependent on AKDS being licensed.

A new User Interface (UI) will be provided (similar to the AKDS UI) which will allow Member Banks to generate the required RSA key pair for their SPAN2 switch. It will also allow them to enter the SAMA public key into their database and provide facilities for the RSA key data in the database to be viewed.

Three new commands will be implemented to provide the following functions:

- Generate the random string sequence, which for convenience will be a random DES key
- Generate a digital signature using the Member Bank public key
- Verify a digital signature using the SAMA public key

When the Member Bank originates a logon/logoff request, the SPAN2 interface will first need to generate a random string which has been specified as a DES key by SAMA. Once the random string has been generated the SPAN2 interface will generate a digital signature for the 1804 message using defined signature fields (including the random string).

This enhancement is supported by the Atalla AKB, DEP and Thales HSMs.

Wincor AKDS support of NCR ATMs

This project enhances TSS and BASE24-atm to support the multi-vendor environment where there is Wincor software running on NCR ATMs. Even though both Wincor and NCR utilize digital signatures, the Wincor implementation is slightly different from NCR's which necessitated the change.

AKDS PCI enhancement

This enhancement adds support to AKDS to be compliant with Payment Card Industry (PCI) standards. Enhancements include:

- Generation of Random number via hardware (previously done in software)
- Logging of event messages to comply with TG-3 requirements (i.e., when a cryptographic key is created or used during processing)
- Extraction and storage of certificate validity dates (certificate based schemes only)
- Normalization of the Host Public Key data sources from a primary and secondary file to a single file

Changes were made in the TSECI code to support these enhancements. Multiple HSM needed to be modified for support along with the Host Public Key and Channel Public Key Data Source Classes. UI changes were made to support a new KEYGEN UI and a conversion program was developed to convert records to the new format of the Host Public Key Security Data Source.

This enhancement is supported by the Atalla AKB, DEP, and Thales HSMs.

AKDS Capacity Report

The AKDS Capacity Report sequentially reads through the Channel Public Key Security data source (chnpkd). This enhancement implements modifications to make the report format for the AKDS Capacity Report and the Transaction Based Pricing Report consistent.

EMV CAP support uplift

This enhancement to Transaction Security Services will facilitate MasterCard's CAP authentication program and Visa's equivalent DPA. The CAP and DPA programs facilitate consumer authentication for card not present channels such as Internet banking using EMV-based smart cards to generate one-time passwords.

Metadata external memory tables

Metadata external memory tables are used to access CONFCSV and MDBCSV information from memory rather than from disk. CONFEMT is the external memory table for CONFCSV information. MDBEMT is the external memory table for MDBCSV information. By accessing metadata from memory rather than from disk, performance is improved and new versions of data sources can be implemented in TSS processes on a rolling basis (one at a time) without incurring a complete outage of the entire TSS system.

Version 08.2 enhancements

MetaMan utility

A new MetaMan Utility was created in this enhancement. This new utility is a full replacement of the preceding SQL-based MetaMan application, and it eliminates the need for a dedicated MS-SQL server.

This utility is run on the target platform similar to the way DALCI is run. This utility has a command to include files which contain table definitions and assign statements. This utility also has a command to generate configuration files, such as CONFCSV, MDBCSV, metadata headers, and Script files.

Reorganize non-C++ files

The purpose of this enhancement is to reorganize non C++ files. These include the following:

- DALCI Insert/Update Scripts (Events and Event Documentation)
- DALCI default configuration files (ISO field, ISO message, Action Code, ATM default configuration)

All the DALCI obey files will be moved from the system\db directory into each component's newly created DALCI directory. This allows for the distribution of configuration information based on the components that are actually in a customer's system. For TSS customers this means that only a subset of configuration information needs to be delivered and installed. In particular, only the Events and Event Documentation required by the TSS application will now be installed, rather than all events for the entire BASE24-eps application.

DALCI enhancements

DALCI insert obey files will be changed from deleting a record (ignoring not found error) followed by an insert, to just insert (ignoring duplicate error). Changing the DALCI insert obey files preserves any modifications made by the customer. Only do delete followed by insert if the record has been updated in product, not just added by product, or just do an update on the record without an insert, assuming the insert has already occurred.

Release and support strategy

An improved product release and support methodology uses a consistent and simplified mechanism to build, deliver, and implement fixes, mandates, and CSMs. This new release and support methodology reduces a customer's dependence on ACI in the development of fixes and CSMs and allows our customers to be proactive in meeting their own needs.

Service pack implementation

A new service pack strategy has been implemented to facilitate a repeatable mechanism to deliver product software of high quality on a consistent basis. With this new methodology, all product fixes will be delivered via a service pack. A service pack is defined as a collection of fixes that have been bundled for delivery to customers as a single installable package. It includes regression testing of all patches and documentation in the form of release notes and installation guides.

Once a BASE24-eps or TSS Release/Version is available as a supported version, service packs will be scheduled and released containing critical and serious core code fixes. No additional core code will be included in these service packs. Version independent packages within a service pack will contain critical, serious, and non-critical fixes and enhancements (e.g., mandates).

BASE24-eps/TSS service packs will be released based on a pre-determined time frame, approximately every 3-4 months, with a goal to coincide with the mandate delivery schedule for Visa and MasterCard. As part of the service pack strategy, the standard support of a BASE24-eps/TSS version has been extended from 2 years to 3 years. Service packs will be delivered on all supported versions.

Patch implementation

A mechanism has been provided for customers to apply a critical patch to their system in situations where installing a full service pack would potentially introduce an unacceptable level of risk to the customer, or if the timing of the next service pack is not acceptable to the customer. Patches will be developed using the CSM methodology (refer to the ***BASE24-eps Customization guide***).

ACI will only develop patches for critical and serious problems. Fixes for non-critical problems for any component other than an interface will be included in the next BASE24-eps/TSS version. Fixes for interface non-critical problems will be included in the next interface Service Pack.

By using the CSM methodology to create fixes, you will be able to develop your own patches, ensuring that issues are resolved as quickly as possible. Of course, ACI appreciates it when our customers share any patches they have created so we can make them available to other customers and include them in the product stream.

Version/Service Pack/Patch (VSP) registration

The VSP registration process creates a way for you to track the various levels of software installed at your site (i.e., Version level, Service Pack level for each component, patches for components, and CSMs for components).

These details will be available by running a specific process control command and will be provided by the customer as part of a problem report. ACI will use this information as part of problem resolution.

Common development environment

ACI will provide a common development environment for use by customers and ACI personnel to enable development of custom software modifications (CSMs) and/or patches.

Platform specific requirements are defined in the ***BASE24-eps Release 1.0 Version 08.2 hardware/software requirements document***.

Binary compatibility – TDE rework and CSM factory enablement

To facilitate binary compatibility, all BASE24-eps C++ code must allow for changes via the CSM Methodology. All objects should be created via a factory mechanism. This will allow all customer specific changes to be done using the CSM Methodology as well as patches, which accomplishes a main objective of keeping CSMs and patches out of inline code changes to product. Most product code already follows this convention.

Enhanced user ID and password encryption

The password associated with a USERID is currently stored after being hashed using SHA-1. While this may meet the PCI DSS requirements, as PCI DSS does not specify the method to encrypt passwords, information stored in this form is open to dictionary (known password) attacks. It is industry best practice to use a salted hash method to mitigate such an attack. With the recently published problems with SHA-1, use of a stronger HASH, such as SHA-256 is recommended. This enhancement increases the password protection from SHA-1 to a salted SHA-256 method.

The support of salted SHA-256 hashing is included in JSF and UIF 07.4. BASE24-eps has been upgraded to this release and enabled for this hashing method.

Version 09.2 enhancements

Enhanced user security

USEC 1.2 has been implemented for BASE24-eps 09.2. The user.security.version property has been set to USEC 1.2 in the APPLCNFG table. This implementation option focuses on improving new user configuration times. It requires a couple of new database tables and improves the filter configuration methodology. Rather than assigning filters to permissions, filters can be assigned to a user or a role.

For customers upgrading to new releases of BASE24-eps, it is recommended that User Security be reconfigured from scratch; nonetheless, when USEC 1.2 is started, it will automatically migrate the customer USEC data from using the old USEC 1.1 table configuration to the new USEC 1.2 table configuration.

Upgrade ACI Web Access Services (WAS) to latest Tomcat versions

Prior to version 09.2, ACI Web Access Services (WAS) was delivered with Tomcat 5.0. Tomcat 5.0 has several documented security vulnerabilities. With this release, depending upon the platform, WAS will be delivered with a newer version of Tomcat.

For HP NonStop S-Series, WAS will be delivered with Tomcat 5.0, which is the final version supporting Java 1.4.2 (the maximum version of Java supported on that platform).

Version 11.1 enhancements

Interac EMV Flash for BASE24 TSS

Interac (Canadian Interchange) has piloted a new EMV initiative known as Interac Flash which allows EMV transactions to be performed in contact and contactless modes to noticeably shorten the transaction cycles for small amount payments. Acquirers and Issuers within Canada have the option to support transactions using these new EMV cards.

Interac already supports EMV; however, the new dual interface cards will use a new Cryptogram Version Number: 85. Acquirers and Issuers may have to support both formats of EMV.

Most ACI customers in Canada run a BASE24 system using BASE24-eps TSS for HSM related calls. The new security capabilities will be used by BASE24 in support of POS acquiring. TSS has been enhanced to support the new Cryptogram Version Number (CVN) 85 for Application Cryptogram (AC) validation, Authorization Response Cryptogram (ARPC) generation, and Secure Messaging for integrity and confidentiality for Interac Flash cards.

This enhancement provides the cryptographic support for Interac Flash for use by BASE24:

HP Nonstop Platform

Atalla AKB

NCR enhanced Remote Key Loading (RKL) for BASE24 TSS

This enhancement adds Transaction Security cryptographic calls to support the NCR enhanced Remote Key Loading (RKL) Protocol scheme. The new security capabilities will be used by BASE24 in support of NCR ATM device acquiring. Transaction Security continues to support the NCR basic scheme in addition to the enhanced scheme.

TSS user interface appearance

TSS users will notice a difference in the appearance of the Channel Security Configuration and Interface Security Configuration UIs. These changes are the result of enhancements that will be implemented for BASE24-eps. The following summarizes the changes and specifies how TSS users should configure the new fields.

Channel security configuration

1. The top section of the UI contains a new field called "Key Hierarchy". This should remain set to the default setting (two levels).
2. The following tabs contain "Current" keys and "Future" keys. The Future keys are not used by TSS.
 - PIN Information
 - MAC Information
 - Data Encryption Exchange
 - Data Encryption Key Info

Interface security configuration

1. The top section of the UI contains a new field called "Key Hierarchy". This should remain set to the default setting (two levels).
2. The top section of the UI contains a check box called "Series". This should remain unchecked.
3. The Exchange Information tab has two sub-tabs, one for PIN Exchange keys and another for MAC Exchange keys. Each sub-tab contains the following keys:
 - Import (C)
 - Export (C)
 - Import (F)
 - Export (F)

The (C) indicates "Current", and the (F) indicates "Future." TSS only uses the Current keys.
4. The Data Encryption Exchange Information tab contains the following keys:
 - Import (C)
 - Export (C)

- Import (F)
- Export (F)

The (C) indicates "Current", and the (F) indicates "Future." TSS only uses the Current keys.

5. The PIN Information, MAC Information, and Data Encryption Key Info tabs contains the following keys:
 - Current 1
 - Current 2
 - Future
6. The following tabs should be disabled when the Key Hierarchy field is set to "two levels". The information on these tabs is not used by TSS.
 - Master Info
 - Level 2 Import Master Info

ACI desktop uplift to Java, version 6

The BASE24-eps 11.1 ACI desktop has been enhanced to utilize and deploy with Java, version 6 using the Java Development Kit (JDK) 1.6 release. This will provide a more efficient operating environment and will provide new possibilities for future enhancements.

With the Java, version 6 uplift, the latest AJI (AJSF and AJPF) framework has been updated to utilize the JDK 1.6 release; however, the target object JDK version for the server components is 1.4 (ESWeb will still only be valid up to a 1.4 JDK).

ACI desktop supports latest Microsoft O/S

BASE24-eps software deliveries requested from ACI Electronic Distribution (ED) can now use a desktop running Microsoft Windows 7.

The BASE24-eps 11.1 ACI desktop can now be installed and supported on the following Microsoft Windows platforms:

- Windows XP
- Windows Vista (32-bit)
- Windows Vista (64-bit)
- Windows 7 (32-bit)
- Windows 7 (64-bit)

ACI Web Access Services (WAS) and Tomcat upgrade to latest versions

For HP NonStop S-Series, ACI WAS will be delivered with Apache Tomcat 5.5, which is the final version supporting Java 1.4.2 (the maximum version of Java supported on that platform).

2: Database impacts

Database impacts for 03.1

New files

The following database files are new with this release of BASE24-eps TSS:

- EPP Serial Number data source (EPNUMD)
This new file extends the functionality of the Automated Key Distribution System (AKDS) add-on product. It provides the ability for validating and auditing the use of specific Encrypting PIN Pad (EPP) devices within a network. It is used to track where EPPs are currently installed (i.e., the Terminal ID where the EPP is located) as well as the last Date and Time that the EPP registered with the system.
- Identikey data source (IDNTD)
This new file is used to support the Identikey method of PIN verification. It contains the information necessary to perform PIN verification and PVV generation using Atalla's Identikey method.
- Identikey OLTP data source (IDNTOD)
This new file contains the information used by the online TSS processes to support the Identikey method of PIN verification. It is loaded from information contained in the Identikey data source.

Modified files

The following existing database files have been modified with this release of the Transaction Security Services subsystem:

- Channel Security data source (CSECD)
This existing file has been modified to provide the additional security information required to support the data encryption add-on product. Additional Exchange Key, Data Encryption Key, and Initialization Vector information has been added. As a result, existing 02.4 versions of this file require a conversion as part of the implementation of version 03.1 of BASE24-eps TSS.
- Interchange Security data source (ISECD)
As with the Channel Security data source, this existing file has been modified to provide the additional security information required to support the data encryption add-on product. Additional Exchange Key, Data Encryption Key, and Initialization Vector information has been added. As a result, existing 02.4 versions of this file require a conversion as part of the implementation of version 03.1 of BASE24-eps TSS.

Corresponding changes have also been made to the online version (OLTP) of this file (ISECOD).

Other files

Several additional standard BASE24-eps TSS database files, while not changing structurally, do require updated or new information. These include the following:

- Event data source (EVENTD)
Several new log messages were created as part of this release. Consequently, the Event data source must be updated. The EVENT DALCI script is provided that includes this new information.
- Event Document data source (EVDOD)
Several new log messages were created as part of this release. Consequently, the Event Document data source must be updated. An EVDOD DALCI script is provided that includes this new information.

Database impacts for 03.2

New files

The following database file is new with this release of BASE24-eps TSS:

- NCR PIN Verification data source (NCRD)
This is a new data source that contains PVK keys for NCR PIN verification.

Modified files

The following existing database file has been modified with this release of BASE24-eps TSS:

- Interchange Security data source (ISECD)
This existing file has been modified to provide the additional security information required to support the Dutch Security enhancement.
Corresponding changes have also been made to the online version (OLTP) of this file (ISECOD).

Other files

Several additional standard BASE24-eps TSS database files, while not changing structurally, do require updated or new information. These include the following:

- Event data source (EVENTD)
Several new event messages were created as part of this release. Consequently, the Event data source must be updated. The EVENT DALCI script is provided that includes this new information.

- Event Document data source (EVDODCD)

Several new event messages were created as part of this release. Consequently, the Event Document data source must be updated. An EVDOD DALCI script is provided that includes this new information.

Database impacts for 03.3

New files

New files – BASE24-eps TSS

- Application Configuration (APPLCNFG)

This file contains configuration information for the ESWeb process.

- External Connection (EXTRCNCT)

This file contains connection information for applications that the ESWeb process needs to communicate with. An example of this would be the XML server.

- Process Definitions (PROCESS)

This file is used with the External Connection file to name the different connections.

New files – User Security Server (USEC)

The following database file is new with this release of the User Security database:

- IP to DNS Name Mapping (HOSTS)

This file contains IP to DNS name mappings.

Modified files

Modified files – BASE24-eps TSS

The following existing database files have been modified with this release of BASE24-eps TSS:

- Channel Security data source (CSECD)

This existing file has been modified to provide the additional security information required to support the Interac Key Management enhancement.

- CONFCSV

Modifications were made to the CONFCSV to place a 16 character limit on Assign Short Names.

Modified files – User Security Server (USEC)

The following existing database files have been modified with this release of the User Security database:

- Filter Field Values (FLDVLS)
This existing file has been modified with the creation of a new alternate key field resulting in the creation of a new alternate key file (FLDVLS0).
- Filters (FLTERS)
This existing file has been modified with the creation of two new alternate key fields resulting in the creation of two new alternate key files (FLTERS0 and FLTERS1).
- Filter Field Value Link (FLXFLV)
This existing file has been modified with the creation of a new alternate key field resulting in the creation of a new alternate key file (FLXFLV0).
- Permission IDs (PERMS)
This existing file has been modified with the creation of two new alternate key fields resulting in the creation of two new alternate key files (PERMS0 and PERMS1).
- Permission Groups (PRMGRP)
This existing file has been modified with the creation of a new alternate key field resulting in the creation of a new alternate key file (PRMGRP0).
- Security Group (SECGRP)
This existing file has been modified with the creation of a new alternate key field resulting in the creation of a new alternate key file (SECGRP0).
- Tasks (TASKS)
This existing file has been modified with the creation of a new alternate key field resulting in the creation of a new alternate key file (TASKS0).
- Users (USERS)
This existing file has been modified with the creation of a new alternate key field resulting in the creation of a new alternate key file (USERS0).

Other files

Several additional standard BASE24-eps TSS database files, while not changing structurally, do require updated or new information. These include the following:

- Event data source (EVENTD)
Several new event messages were created as part of this release. Consequently, the Event data source must be updated. The EVENT DALCI script is provided that includes this new information.

- Event Document data source (EVDOD)

Several new event messages were created as part of this release. Consequently, the Event Document data source must be updated. An EVDOD DALCI script is provided that includes this new information.

Database impacts for 04.1

New files

New files – BASE24-eps TSS

The following database files are new with this version of BASE24-eps TSS:

- Derivation Key (DRVKD) data source
This new data source contains the DUKPT derivation keys. A corresponding OLTP file (DRVKOD) is also required for online processing.
- User Audit Log (UALOGD)
The User Audit Log table is used to store the User Interface audit records. This table stores information on user interactions with an application's data sources. The data stored in the audit entry detail column in the table is in XML format.

Modified files

Modified files – BASE24-eps TSS

The following existing database files have been modified for this version of BASE24-eps TSS:

- EMV Security data source (EMVSD)
This existing file has been modified to provide the fields required to allow the specification of the Derived Key Index field. Corresponding changes have also been made to the online version (OLTP) of this file (EMVSOD).
- BASE24 KEYF
This existing BASE24 file has been modified to include the additional counter information required for Interac Key Management support.

Modified files – Java Server

The following existing database files have been modified with this release of the Java Server database:

- APPLCNFG (APPLD)

This file is used by the Java Server environment and contains application configuration properties used during UI processing. A change was made to this file to increase a field size.

- EXTRCNCT (EXTCND)

This file is used by the Java Server environment and contains external connection configuration data that identifies connection properties between the Java Server and C++ and other third-party environments. A change was made to this file to increase a field size.

Deleted files

Deleted files – Java Server

The Node controller was removed in 04.1 so the RUNNC and KILLNC files no longer exist.

Other files

Several additional standard TSS database files, while not changing structurally, do require updated or new information. These include the following:

- Event data source (EVENTD)

Several new event messages were created as part of this release. Consequently, the Event data source must be updated. The EVENT DALCI script is provided that includes any new or updated information.

- Event Document data source (EVDOD)

Several new event messages were created as part of this release. Consequently, the Event Document data source must be updated. An EVDOD DALCI script is provided that includes any new or updated information.

Database impacts for 04.4

New files

New files – BASE24-eps TSS

The following database files are new with this version of BASE24-eps TSS:

- UI Contingency Audit Store and Forward (UJSAFD)
- UI Contingency Audit Store and Forward Outstanding (UJSFOD)

Modified files

- CONFCSV

Modifications were made to the CONFCSV to include the User Security entries.

- MDBCSV

Modifications were made to the MDBCSV to include the User Security entries.

Renamed files

With version 04.4, the USEC and BASE24-eps TSS CONFCSV and MDBCSV files were merged to reduce duplication of configuration information. To ensure that file names between the two applications were not duplicated, all physical filenames in the USEC database were renamed to contain a starting u character (i.e., uusers). If the previous file name was six characters in length, the final character was dropped to ensure that the maximum file name length of six characters was enforced. The exception to the six-character limit is alternate key files that contain the primary file name appended with a single-digit alternate key number.

Other files

Several additional standard BASE24-eps TSS database files, while not changing structurally, do require updated or new information. These include the following:

- Event data source (EVENTD)

Several new event messages were created as part of this release. Consequently, the Event data source must be updated. The EVENT DALCI script is provided that includes any new or updated information.

- Event Document data source (EVDOD)

Several new event messages were created as part of this release. Consequently, the Event Document data source must be updated. An EVDOD DALCI script is provided that includes any new or updated information.

Database impacts for 05.2

New files

None.

Modified files

The following standard BASE24-eps TSS database files were modified for this release:

- SECD CD
Modifications were made to the SECD CD to include the new parameters for Hardware Model, Protocol/Interface, Key Scheme, Firmware Version, Maximum Buffer Length, and Reset Timer Limit.
- CONF CSV
Modifications were made to the CONF CSV to update the Security Device Configuration entry.
- MDB CSV
Modifications were made to the MDB CSV to update the Security Device Configuration entries.

Renamed files

None.

Other files

Several additional standard BASE24-eps TSS database files, while not changing structurally, do require updated or new information. These include the following:

- Event data source (EVENT D)
Several new event messages were created as part of this release. Consequently, the Event data source must be updated. The EVENT DALCI script is provided that includes any new or updated information.
- Event Document data source (EVD O D)
Several new event messages were created as part of this release. Consequently, the Event Document data source must be updated. An EVD O C DALCI script is provided that includes any new or updated information.

Database impacts for 05.4

New files

None.

Modified files

The following standard BASE24-eps TSS database file was modified for this release:

- CONFCSV

Modifications were made to the CONFCSV to add the AKDS_CAPACITY_REPORT entry.

Renamed files

None.

Other files

None.

Database impacts for 06.2

New files

None.

Modified files

None.

Renamed files

None.

Other files

None.

Database impacts for 06.4

New files

The following new files have been added with this release of BASE24-eps TSS:

- Dynamic Card Verification (DCRDVD)

This data source is used to hold the keys and processing parameters required to support the processing for the dCVV and CVC3 (Dynamic Card Verification) enhancement.

- Dynamic Card Verification OLTP (DCRDOD)

This is the OLTP data source built from the DCRDVD base file.

Modified files

Modified files – BASE24-eps TSS

The following existing database files have been modified with this release of BASE24-eps TSS:

- MDBC SV

Modifications were made to the MDBC SV to change all fields used to carry keys from fixed length ASCII strings to fixed length binary data. This change impacts the following security related files:

Card Verification (CRDVD)

Card Verification OLTP (CRDVOD)

Channel Public Key Security (CHNPKD)

Channel Security (CSECD)

Diebold Number Table (DNTD)

Diebold Number Table OLTP (DNTOD)

Derivation Key (DRVKD)

Derivation Key OLTP (DRVKOD)

EMV Security (EMVSD)

EMV Security OLTP (EMVSOD)

Host Public Key Security Primary (HSPKPD)

Host Public Key Security Secondary (HSPKSD)

IBM DES (IDESD)

IBM DES OLTP (IDESOD)

Interchange Security (ISECD)

Interchange Security OLTP (ISECOD)

NCR PIN Verification (NCRD)

NCR PIN Verification OLTP (NCROD)

Transport Key (TRNSD)

- VISA PVV (VPVVD)

VISA PVV OLTP (VPVVOD)

No conversion of existing data sources is required as modifications were made to code to handle the change. This change will impact how DALCI Insertform commands are generated. Key fields will now be generated in HEX format rather than ASCII. Existing DALCI scripts which contain key fields in ASCII can continue to be used without modification.

- EMV Security (EMVSD) and EMV Security OLTP (EMVSOD)
Modifications were made to these data sources to add the new Crypto Version Number field required to support CCD processing.
- BASE24 KEYF (KEYF)
Fields were added to this data source for BASE24 HISO Full Message Encryption support.
- Channel Public Key Security (CHNPKD)
Fields were added to this data source for future enhancements.

Renamed files

None.

Other files

None.

Database impacts for 07.4

New files

The following new files have been added with this release of BASE24-eps TSS:

- Host Public Key Security (HSPKD)
This new file is a normalization of the Host Public Key Security Primary (HSPKPD) and Host Public Key Security Secondary (HSPKSD) data sources. A program is provided to allow conversion of existing customer files to the new format file.
In addition to providing for all the existing fields, the new data source also included new fields to contain certificate validity dates where applicable.
- Chip Authentication (CAD)
This new file contains the information necessary to perform Chip Authentication (CAP/DPA) functionality.
- Chip Authentication OLTP (CAOD)
This is the external memory table (OLTP) that corresponds to the Chip Authentication data source.
- CONFCSV External Memory Table (CONFEMT)
This is the external memory table that corresponds to the CONFCSV.
- MDBCSV External Memory Table (MDBEMT)
This is the external memory table that corresponds to the MDBCSV.

Modified files

Modified files – BASE24-eps TSS

The following existing database file has been modified with this release of BASE24-eps TSS:

- Channel Public Key Security (CHNPKD)

Fields were added to this data source to contain the following information:

- EPP and Host generated random nonce values.
- EPP Serial Number

Note: Validity date fields added to this data source in 06.4 will now be utilized.

Renamed files

None.

Other files

None.

Database impacts for 08.2

New files

The following new files have been added with this release of BASE24-eps TSS:

- APPLDSRC or Application Data Source (APSRCD)

This new file defines application IDs that are related to a specific data source (alternate database).

- Compiler Message Documentation (CMSGD)

This new file contains one record for each message which is generated by the script compiler for errors and warnings. The record contains the severity of the message, the message text, and message detail.

- EMV Card Verification Results (CVR) (CVRD)

This new file defines the Card Verification Results checks that are to be performed during routing and/or authorization of EMV transactions and the actions to be taken for a specific CVR byte/bit setting. The CVR is a proprietary EMV data element (included in the Issuer Application Data). Not used by TSS at this time.

- CVRD0

The alternate key file for the EMV Card Verification Results file. Not used by TSS at this time.

- EMV Card Verification Results OLTP (CVROD)
This is the external memory table (OLTP) that corresponds to the EMV Card Verification Results file. Not used by TSS at this time.
- EMV Card Verification Results (CVR) Status by Order OLTP (CVROOD)
This new file allows look-ups by search order by CVR Profile/EMV Format/Table ID combination. Not used by TSS at this time.
- Event Log (EVTLOG)
This new file is used for event consolidation and perusal.
- HSMCNFG or HSM Configuration (HSMCNF)
This new file contains host security module configuration.
- Instrument Type (ITYPD)
This new file contains the instrument types and is used by the UI to maintain the instrument types and descriptions. Not used by TSS at this time.
- Key Token (KTD)
This new file provides a means of storing non-standard key data. With this release, it is only used with the IBM CryptoExpress2 implementation that is applicable only to the IBM z/OS environment. With standard TSS installations, an empty file will be created but never used by the TSS application.
- MSGCTXT or Message Context (MSGCTX)
This new file contains context-sensitive data.
- Exported Operator Documentation (OPDOCD)
This new file contains information about the registered exported operators on a system. The operator descriptions, prerequisites, return values, and parameter values are configured in this table.
- Exported Operator Documentation OLTP (OPDOCD0)
The alternate key file for the Exported Operator Documentation data source.
- EMV Online Reason Code (ROCD)
This new file defines the Reason Code Online Code checks that are to be performed during routing and/or authorization of EMV transactions and the actions to be taken for a specific code. Not used by TSS at this time.
- EMV Online Reason Code Description (ROCDD)
This new file contains the description for a reason online code value. Not used by TSS at this time.
- EMV Online Reason Code OLTP (ROCOD)
This is the external memory table (OLTP) that corresponds to the EMV Reason Online Code file. Not used by TSS at this time.

- **Script Configuration (SCPTCD)**
This new file contains one record for each executable script in the system. An executable script may be a high level script or a subscript. The script compiler updates the parameter types and return values. Not used by TSS at this time.
- **Script Configuration OLTP (SCPTOD)**
This is the external memory table (OLTP) that corresponds to the Script Configuration data source. Not used by TSS at this time.
- **Script (SCRPTD)**
This new file contains one record for each compiled script statement. The Script Component writes records to this table when compiling scripts. Not used by TSS at this time.
- **Script Source (SCSRCD)**
This new file contains one record for each line of a script. All lines together define the flow of a script. The script source is compiled, and the output is placed in the Script data source and updated in the Script Configuration data source. Not used by TSS at this time.
- **EMV Terminal Verification Results (TVRD)**
This new file defines the Terminal Verification Results checks that are to be performed during routing and/or authorization of EMV transactions and the actions to be taken for a specific TVR byte/bit setting. Not used by TSS at this time.
- **TVRD0**
The alternate key file for the EMV Terminal Verification Results file. Not used by TSS at this time.
- **EMV Terminal Verification Results OLTP (TVROD)**
This is the external memory table (OLTP) that corresponds to the EMV Terminal Verification Results file. Not used by TSS at this time.
- **EMV Terminal Verification Results (TVR) Status by Order OLTP (TVROOD)**
This new file allows look-ups by search order by TVR Profile/Table ID combination. Not used by TSS at this time.
- **ACTSESS2 or Active Sessions (UACTS2)**
This new file contains any existing active user session (authenticated users) with extended session strength.
- **FLTROLE or Filters to Roles Relationship Table (UFLXRL)**
This new file contains relationships of filters to roles. For future use.
- **USRXFILT or Users to Filter Relationship Table (UUSRXF)**
This new file is used to map users to filter groups.
- **Versions (VERSD)**
This new file contains resource initialization map versions of XML files that are used to seed/initialize application-specific data into a database.

TSSMETA (where TSS is your System Prefix that you set on your Pre-Install Checklist and Worksheet) is a new subvolume which will contain the Metadata for installed components.

The following new files have been added to the TSSCNFG subvolume (where TSS is your System Prefix that you set on your Pre-Install Checklist and Worksheet):

- ACILMN
This new file creates the MetaMan utility executable.
- ACILSCPT
Part of SCRIPT component. Not needed by TSS at this time.
- ENVVARS
Generated by the Install program. Contains variables used by OSS during install.
- METAMN
TACL Macro to run MetaMan utility.
- MNM
Output file created by ACILMN.
- MNNL
Output file created by ACILMN.
- SCPTENSC
Part of SCRIPT component. Not needed by TSS at this time.
- SCPTM
Part of SCRIPT component. Not needed by TSS at this time.
- SCPTNL
Part of SCRIPT component. Not needed by TSS at this time.
- New OSS file {ESHOME}/envvars
This new OSS file was created to resolve a TACL params space issue which caused ESSETUP to fail. It replaces ESWEBPRM and ESWEBSET.

Modified files

None.

Renamed files

None.

Other files

The CACCAT, CRDCAT, and PBLCAT files will no longer be created during a TSS install. They were moved to BASE24-eps-specific directories and are not used by TSS customers.

The ESWEBPRM and ESWEBSET files have been replaced by the new OSS ENVVARS file.

Because of the changed MetaMan utility as of 08.2, a new KEYFD file is being created during install and also the execution of ESFIX. This file corresponds to the KEYF file (the DDL layout of the BASE24 KEYF.) You are instructed on how to work with this new file in section 6.

Database impacts for 09.2

New files

The following database files are new with this release of BASE24-eps TSS:

- ZKA Key Generation Keys Data Source (ZKGKDS)
This new file contains the Key Generation Keys needed in German Security Support. This file is not used by TSS at this time.
- Dual PVN PIN Verification Data Source (DPVND)
This new file is used for storage and retrieval of Key Generation Keys in the processing of Dual PVN PIN Verification requests. This file is not used by TSS at this time.
- Dual PVN PIN Verification OLTP Data Source (DPVNOD)
This is the external memory table (OLTP) that corresponds to the Dual PVN PIN Verification file. This file is not used by TSS at this time.
- EMV Issuer Authentication Data Source (EIADD)
This new file is used for configuration of EMV Card Status Update (CSU) and Proprietary Authentication Data (PAD) based on the Internal Action Code and the Method. This file is not used by TSS at this time.
- EMV Issuer Authentication OLTP Data Source (EIADOD)
This is the external memory table (OLTP) that corresponds to the EMV Issuer Authentication file. This file is not used by TSS at this time.
- Web Application Options (WBOPTD)
This new file stores the default user preferences options for the web application.
- Web Applications (WEBAPD)
This new file is used to create a jump page (Other Applications web page) for single sign-on (SSO) access to multiple web applications.
- Web Application User Options (WBUSRD)
This new file stores the user preferences selected by the web application's users on the User Preferences page.

Modified files

The following database files are modified with this release of BASE24-eps TSS:

- Channel Security Data Source File (CSECD)

Fields were added to this data source to contain the following information. None of them is used by TSS.

- ☐ Inbound MAC Key Type
- ☐ Outbound MAC Key Type
- ☐ Inbound TEK Key Type
- ☐ Outbound TEK Key Type
- ☐ Terminal Encryption Key Header
- ☐ Terminal Encryption Key or Terminal Encryption Key Receive Key Value
- ☐ Terminal Encryption Key MAC Value
- ☐ Terminal Encryption Key Verification Value
- ☐ Terminal Encryption Key Length
- ☐ Previous Terminal Encryption Key Header
- ☐ Previous Terminal Encryption Key or Previous Encryption Key Receive Key Value
- ☐ Previous Terminal Encryption Key MAC Value
- ☐ Previous Terminal Encryption Key Verification Value
- ☐ Cross Acquirer Key Exchange Key Header
- ☐ Cross Acquirer Key Exchange Key Value
- ☐ Cross Acquirer Key Exchange Key MAC Value
- ☐ Cross Acquirer Key Exchange Key Verification Value
- ☐ Cross Acquirer Key Exchange Key Length
- ☐ Acquirer Initialization Key Header
- ☐ Acquirer Initialization Key Value
- ☐ Acquirer Initialization Key MAC Value
- ☐ Acquirer Initialization Key Verification Value
- ☐ Acquirer Initialization Key Length
- ☐ Initial MAC Key Header
- ☐ Initial MAC Key Value
- ☐ Initial MAC Key MAC Value
- ☐ Initial MAC Key Verification Value
- ☐ Initial MAC Key Length
- ☐ Terminal Master Key 1 Header

- ☐ Terminal Master Key 1 Value
- ☐ Terminal Master Key 1 MAC Value
- ☐ Terminal Master Key 1 Verification Value
- ☐ Terminal Master Key 1 Length
- ☐ Previous Terminal Master Key 1 Header
- ☐ Previous Terminal Master Key 1 Value
- ☐ Previous Terminal Master Key 1 MAC Value
- ☐ Previous Terminal Master Key 1 Verification Value
- ☐ Terminal Master Key 2 Header
- ☐ Terminal Master Key 2 Value
- ☐ Terminal Master Key 2 MAC Value
- ☐ Terminal Master Key 2 Verification Value
- ☐ Previous Terminal Master Key 2 Header
- ☐ Previous Terminal Master Key 2 Value
- ☐ Previous Terminal Master Key 2 MAC Value
- ☐ Previous Terminal Master Key 2 Verification Value
- ☐ Previous Terminal Master Key 2 Length
- ☐ ZKA Key Derivation Flag
- ☐ Key Generation Number (KGN) for the derivation key
- ☐ Key Version Number (KVN) for the Derivation Key
- ☐ Key Table Number (KTN) for the Derivation Key
- ☐ Derivation depth of the Derivation key
- ☐ Derivation data used to derive the Terminal Specific key
- Interchange Security Data Source File (ISECD)

Fields were added to this data source to contain the following information. None of them is used by TSS.

 - ☐ Inbound MAC Key Type
 - ☐ Outbound MAC Key Type
 - ☐ Key type of the import transport key
 - ☐ Key type of the export transport key
 - ☐ Inbound ZEK Key Type
 - ☐ Outbound ZEK Key Type
 - ☐ Input Key Type used to indicate the use of ZMK or KEK as input for some Thales AS2805 commands
 - ☐ ZKA Key Derivation Flag

- ❑ Key Generation Number (KGN) for the derivation key
- ❑ Key Version Number (KVN) for the Derivation Key
- ❑ Key Table Number (KTN) for the Derivation Key
- ❑ Derivation depth of the Derivation key
- ❑ Derivation data used to derive the Terminal Specific key
- ❑ Pending inbound PIN KEK header
- ❑ Pending inbound PIN KEK key
- ❑ Pending inbound PIN KEK MAC
- ❑ Pending inbound PIN KEK check digits
- ❑ Pending inbound MAC KEK header
- ❑ Pending inbound MAC KEK key
- ❑ Pending inbound MAC KEK MAC
- ❑ Pending inbound MAC KEK check digits
- ❑ Pending outbound Data KEK header
- ❑ Pending outbound Data KEK key
- ❑ Pending outbound Data KEK MAC
- ❑ Pending outbound Data KEK check digits
- Derivation Key File (DRVKD)
 - The following field was added:
 - ❑ Encryption DES Mode

Renamed files

None.

Other files

None.

Database Impacts for 11.1

New files

The following database files are new with this release of BASE24-eps TSS:

- **Interchange_Security_Future (ISECFD)**
Used to configure the future keys needed for an interface. This data source is a logical continuation of the Interchange Security data source at the record level. All the key characteristics (length, timers, etc.) are kept only in current keys data source (Interchange Security). This file is not used by TSS at this time.
- **Channel_Security_Future (CSECFD)**
Used to configure the future keys needed for a channel. This data source is a logical continuation of the Channel Security data source at the record level. All the key characteristics (length, timers, etc.) are kept only in current keys data source (Channel Security). This file is not used by TSS at this time.

ACI Desktop and ACI Java Infrastructure (AJI) impacts

- **EVNTLOG (EVTLOG)**
A new internal table for the Event Adapter. This table allows for events to be maintained in the database for auditing purposes. There is an associated timer task which can be configured to remove entries that are older than a certain date.

Modified files

The following database files are modified with this release of BASE24-eps TSS:

- **Channel Security Data Source File (CSECD)**
Fields were added to this data source to contain the following information. None of them is used by TSS.
 - ❑ Key Hierarchy
 - ❑ KKT Option
 - ❑ Master 1 Export header
 - ❑ Master 1 Export key
 - ❑ Master 1 Export MAC
 - ❑ Master 1 Export check digits
 - ❑ Master 1 export KKT length
 - ❑ Inbound Data Encryption Key 1 Level 2 header
 - ❑ Inbound Data Encryption Key 1 Level 2 key
 - ❑ Inbound Data Encryption Key 1 Level 2 MAC
 - ❑ Inbound Data Encryption Key 1 Level 2 check digits
 - ❑ Inbound Data Encryption Key 1 Level 2 old header

- ☐ Inbound Data Encryption Key 1 Level 2 old key
- ☐ Inbound Data Encryption Key 1 Level 2 old MAC
- ☐ Inbound Data Encryption Key 1 Level 2 old check digits
- ☐ Future Keys Usage
- INTERCHANGE_SECURITY data source (ISECD) and OLTP (ISECOD)
The following new data elements were added. None of them are used by TSS.
 - ☐ Key Hierarchy
 - ☐ KKT Option
 - ☐ Master 1 Export header
 - ☐ Master 1 Export key
 - ☐ Master 1 Export MAC
 - ☐ Master 1 Export check digits
 - ☐ Master 1 Import header
 - ☐ Master 1 Import key
 - ☐ Master 1 Import MAC
 - ☐ Master 1 Import check digits
 - ☐ Master 1 KKT length
 - ☐ Master 1 Import PIN KEK header
 - ☐ Master 1 Import PIN KEK key
 - ☐ Master 1 Import PIN KEK MAC
 - ☐ Master 1 Import PIN KEK check digits
 - ☐ Master 1 KPT length
 - ☐ Master 1 Import MAC KEK header
 - ☐ Master 1 Import MAC KEK key
 - ☐ Master 1 Import MAC KEK MAC
 - ☐ Master 1 Import MAC KEK check digits
 - ☐ Master 1 KAM length
 - ☐ Master 1 Import Data KEK header
 - ☐ Master 1 Import Data KEK key
 - ☐ Master 1 Import Data KEK MAC
 - ☐ Master 1 Import Data KEK check digits
 - ☐ Master 1 KDT length
 - ☐ Old KEK timer
 - ☐ Future Keys Usage

- Multiple flag
 - Series Enabled flag
 - INTERCHANGE_SECURITY data source (isecd) and OLTP (isecod).
- The following data elements were renamed. None of them are used by TSS.

Previous data element name	Current data element name
kkt_pend_pt_import_hdr	kkt_pt_import_old_hdr
kkt_pend_pt_import_keyx	kkt_pt_import_old_keyx
kkt_pend_pt_import_kam	kkt_pt_import_old_kam
kkt_pend_pt_import_vvk	kkt_pt_import_old_vvk
kkt_pend_am_import_hdr	kkt_am_import_old_hdr
kkt_pend_am_import_keyx	kkt_am_import_old_keyx
kkt_pend_am_import_kam	kkt_am_import_old_kam
kkt_pend_am_import_vvk	kkt_am_import_old_vvk
kkt_pend_dt_import_hdr	kkt_dt_import_old_hdr
kkt_pend_dt_import_keyx	kkt_dt_import_old_keyx
kkt_pend_dt_import_kam	kkt_dt_import_old_kam
kkt_pend_dt_import_vvk	kkt_dt_import_old_vvk
kkt_pend_pt_export_hdr	kkt_pt_export_old_hdr
kkt_pend_pt_export_keyx	kkt_pt_export_old_keyx
kkt_pend_pt_export_kam	kkt_pt_export_old_kam
kkt_pend_pt_export_vvk	kkt_pt_export_old_vvk
kkt_pend_am_export_hdr	kkt_am_export_old_hdr
kkt_pend_am_export_keyx	kkt_am_export_old_keyx
kkt_pend_am_export_kam	kkt_am_export_old_kam
kkt_pend_am_export_vvk	kkt_am_export_old_vvk
kkt_pend_dt_export_hdr	kkt_dt_export_old_hdr

Previous data element name	Current data element name
kkt_pend_dt_export_keyx	kkt_dt_export_old_keyx
kkt_pend_dt_export_kam	kkt_dt_export_old_kam
kkt_pend_dt_export_vvk	kkt_dt_export_old_vvk

ACI Desktop and ACI Java Infrastructure (AJI) impacts

APPLCNFG (APPLD) — The CnfgPropValTx column was increased from 100 bytes to 255 bytes. Fields were added to this data source to contain the following information for internal session management:

- Row Access
- Entry Template
- Validation Template

Filters (UFLTRS) — Fields were added to this data source to contain the following information for internal session management:

- Active
- Read Only

Roles (UROLES) — Fields were added to this data source to contain the following information for internal session management:

- Active
- Read Only

Users (UUSERS) — Fields were added to this data source to contain the following information for internal session management:

- User Active Timestamp
- User Expire Timestamp

Renamed files

None.

Other files

None.

3: Configuration

Configuration for 03.1 enhancements

Dynamic key management

There are no configuration changes between versions 02.4 and 03.1 for this enhancement.

Encrypting PIN Pad (EPP) serial number

The new EPP Serial Number file must be configured to take advantage of this enhancement.

Identikey PIN verification

Configuration of the new Identikey data source (IDNTD) must be accomplished to take advantage of this enhancement.

Data encryption and decryption

Appropriate configuration of the Channel Security data source and the Interchange Security data source must be made to take advantage of this enhancement.

Eracom Mark II security module

Specific information must be configured in the Security Device Configuration data source to take advantage of this enhancement.

EMV offline PIN change

There are no configuration changes between versions 02.4 and 03.1 for the EMV Offline PIN Change.

As a part of the Channel Security and Interchange Security data source conversions, any EMV Offline PIN Change data is converted automatically in these data sources.

New key activation support

There are no configuration changes between versions 02.4 and 03.1 for this enhancement.

Configuration for 03.2 enhancements

Dutch security

Appropriate configuration of the new NCR Authorization data source as well as the additional information in the Interchange Security data source is required to take advantage of this enhancement.

iCVV verification

There are no configuration changes required to support this enhancement.

CVV and CVV2 generation

There are no configuration changes required to support this enhancement.

Configuration for 03.3 enhancements

Enhanced tracing of HSM messages

There are no configuration changes required to support this enhancement.

CAVV verification

There are no configuration changes required to support this enhancement.

Interac key management

Configuration of the new Channel Security fields is required to implement this enhancement.

Configuration for 04.1 enhancements

Derived Unique Key Per Transaction (DUKPT) support

BASE24 customers must run the KEYD conversion program that is provided by ACI. This conversion program moves the Derivation Keys from the BASE24 database to the Derivation Key Data Source in BASE24-eps TSS.

Check digit validation/generation utility

There are no configuration changes required for this enhancement.

Derived Unique Key Per Transaction (DUKPT) MAC support

BASE24 customers must run the KEYD conversion program that is provided by ACI. This conversion program moves the Derivation Keys from the BASE24 database (KEYD) to the Derivation Key Data Source (DRVKD) in BASE24-eps TSS.

EMV Derivation Key Index support

Customers must update the key index values using the User Interface if index values other than 1–24 are used.

If the customer does not currently have 24 crypto keys, they must update the key index fields (through the UI) with spaces for any crypto key rows that do not contain keys.

Interac changes for BASE24 KEYF

The additional information maintained in the KEYF must be manually updated using the UI to utilize this enhancement.

ISO Format 1 PIN block support

The Channel Security and Interchange Security information must be updated to utilize this enhancement.

Configuration for 04.4 enhancements

AMEX Card Security Code (CSC) validation support

The Authentication Profile UI Card tab was modified to add the UI support for the American Express card verification scheme (CSC).

A new field, Verification Algorithm, was added to the Card tab so that the user can select which type of keys are to be entered (Visa/MasterCard or AMEX), and the UI formats the key display based on the verification scheme selected. Visa/MasterCard Verification Algorithm is the default selection.

Special formatting is only required for Atalla non-AKB and Thales HSMs so the new Verification Algorithm value is visible and utilized only for these types of security devices. Existing formatting for AKB format keys required no modification.

Expiration date in card verify message

There are no configuration changes required to support this enhancement.

Encryption type in key generate response

Thales customers must encrypt the decimalization table using a console command and enter the encrypted value in the Decimalization Table field on the IBM DES tab of the Authentication Profile UI. Thales customers must also enable the HSM to support encrypted

decimalization tables using the Configure Security Parameter Encrypted/Plaintext Decimalization Tables. The new Environment parameter, DECIMAL_TBL_ENCRYPT, is required in the Environment data source. This variable needs to be set to the value Y to be able to translate the decimalization table from the old master key to a new master key.

Atalla customers must load the decimalization tables into HSM memory using a console command. The Key Table Location value must be entered in the decimalization table field on the IBM DES tab of the Authentication Profile UI.

Disable public/private key generation (AKDS only)

There are no configuration changes required to support this enhancement.

EMV2000 support

The Authentication Profile UI EMV tab was modified to add the UI support for EMV 2000.

A new field was added to the EMV tab with the following possible selection values. The default value is the first option.

- Branch factor 2; Tree Height 16
- Branch factor 4; Tree Height 8

Eracom AMB hardware security module support

Changes were made to the Authentication, Channel Security, Interface Security, and Security Device Configuration UIs to support a new HSM device type, Eracom AMB. The functionality provided by Eracom AMB differs from the current Eracom Mark II HSM and also has some unique encrypted key display and key entry requirements.

MFK/LMK conversion utility for public keys (AKDS only)

The existing MFK/LMK key conversion utility and UI windows have been modified to include the Host public/secret keys and the Channel Public Keys in the conversion from an old MFK/LMK to a new MFK/LMK.

Delivery of source code documentation

There are no configuration changes required to support this enhancement.

Delivery of TSS in linkable object form

There are no configuration changes required to support this enhancement.

Support for Format 2 files (HP NonStop only)

To create a Format 2 file, the MetaMan output must be changed slightly for FUP to create a Format 2 file. When a system is delivered, all of the database files are created using the

standard Enscribe file type. However, using the instructions in this document, any database file can be created with type Format 2 so that it can exceed the 2 GB file size limit.

The standard TSSFUP file creation script delivered with BASE24-eps TSS deliveries contains all the commands to create the BASE24-eps TSS Database files. To create one or more of the files as a Format 2 file, the following entries (highlighted in **red**) for the target file in the TSSFUP script require modification:

```
Reset
  Set Type      K
  Set Format    2
  Set KeyLen    51
  Set KeyOff    4
  Set Rec       4040
  Set Block     4096
  Set Iblock    4096
  Set Ext       (4, 32)
  Set MaxExtents 100
  Set Audit
Create <data_source_name>
```

NSJ4 support

Customers wanting to support HP NSJ4 also require all XNLIBN and XN30WAX files released with SCR3167 made available 17 September 2004.

Configuration for 05.2 enhancements

CSEC memory enhancement

Two new Environment parameters, CSEC_MEMORY_USE and MAX_CSEC_IN_MEM are used to determine whether the system should maintain a memory-based table of CSEC records and, if so, how many devices this table should support, or whether it should always read the record from disk.

Parameter name	Value range	Default value
CSEC_MEMORY_USE	Y or N	Y
MAX_CSEC_IN_MEM	1-n	20000

For example, if CSEC_MEMORY_USE is set to the value N, CSEC records are always read from disk and a memory table is not employed. If CSEC_MEMORY_USE is set to the value Y, and MAX_CSEC_IN_MEMORY is set to 100, when the 101st CSEC memory is required, the oldest record is removed and replaced with the new record.

CSEC Refresh Process Control command

There are no configuration impacts for the new CSEC REFRESH Process Control command.

Additional security device configuration parameters

- Customers should run the Security Device Configuration conversion program that is provided by ACI to update the SECD CD database file. This program translates the current combination of DEVICE TYPE and DEVICE SUBTYPE into the three new fields HARDWARE MODEL, PROTOCOL/INTERFACE and KEY SCHEME in BASE24-eps TSS. See section 7 for information on how to run the conversion program.
Note: The RSM_MFK_SCHEME Environment parameter is no longer needed after this enhancement is implemented. However, the parameter is required for the Security Device Configuration conversion program. Therefore, the parameter must not be removed until after this conversion program is run.
- Customers must issue an INIT command from the Process Control window for each HSM to populate or repopulate the display-only fields FIRMWARE VERSION and MAXIMUM BUFFER LENGTH for the device on the Security Device Configuration window.
- If the Security Device Configuration PROTOCOL/INTERFACE parameter is set to TCP/IP or TCP/UDP (RAM or URAM), the MSGMODIFY parameter in the RAM/URAM process configuration must be set to 0. This is due to the fact that when the Security Device Configuration within TSEC is set to TCP/IP or TCP/UDP (RAM or URAM), TSEC no longer adds the STX/ETX to the HSM commands. When RAM/URAM is configured with MSGMODIFY set to 1, it expects the HSM commands to be bracketed by STX/ETX and removes them.

Note: This requirement applies only to Thales HSM environments.

HSM automatic reconnect

When an HSM has been marked as out of service, a connection is reattempted automatically at the interval of seconds specified by the new field, RESET TIMER LIMIT, on the Security Device Configuration window, until the HSM is back in service. If the value of RESET TIMER LIMIT is set to 0, there is no automatic attempt to reconnect to the HSM. This must be achieved using the Process Control command ENABLE.

Configuration for 05.4 enhancements

Wincor AKDS support

The following UIs were enhanced to support a Wincor device type (value of 2):

- Certificate Management UI
- Disable Key Gen
- EPP Serial Number Configuration
- Host Public Key Perusal

The UI now allows the user to enter a 16-byte host name.

AKDS Capacity report

There are no configuration impacts for the AKDS Capacity Report enhancement.

Configuration for 06.2 enhancements

APACS MACing support

The Channel Security UI has been enhanced to provide an additional tab for entry of the Initial Key Register. This value must be configured for all devices that will utilize the APACS MACing scheme.

Pseudo 3DES MACing support

The MACing algorithm used is controlled by the value of the MAC Option field on the MAC Information tabs in the Channel Security and Interface Security UIs. Valid values for this field have been modified to include the following:

Single length DES Cipher Block Chaining

3DES Cipher Block Chaining (ISO 9797 MAC Algorithm 1 is used)

Pseudo 3DES Cipher Block Chaining (ISO 9797 MAC Algorithm 3 is used)

Previous versions of TSS only support the third option (Pseudo 3DES) when the configured HSM type was Atalla (either Variant or AKB). This version of TSS adds the Pseudo 3DES option for Thales and Eracom HSMs.

VISA DUKPT MACing

The MACing algorithm used is controlled by the value of the MAC Option field on the MAC Information tab in the Channel Security UI. Valid values for this field have been modified to include the following:

- Old Style Visa DUKPT

This additional MACing algorithm is supported on Atalla HSMs only.

Configuration for 06.4 enhancements

Contactless chip and magnetic card (dCVV and CVC3) support

A new data source (Dynamic Card Verification) has been added as a result of this enhancement. This new data source contains the following information:

- Begin Date

- End Date

- Verification Algorithm (dCVV or CVC3)

- Verification Key

The Authentication UI has been modified to change the Card check box to a drop-down selection box. Valid selections are:

- None (no card verification)

- CVD1/CVD2 (equivalent to current Card Verification)

- Dynamic (new dCVV/CSC3)

- Both (both CVD1/CVD2 and Dynamic).

When either Dynamic or Both is selected, a new Dynamic Card tab is activated.

When configuring Dynamic Card verification, the type of verification being performed (VISA's dCVV or MasterCard Paypass CSC3) must be taken into consideration when generating keys on the Thales HSM. The Thales HSM requires that the verification key be encrypted under different variants of LMK key pair 28-29 (variant 1 for VISA dCVV and variant 9 for MasterCard Paypass CSC3).

CVV1 and CVV2 combined verification support

For BASE24-eps TSS, no additional configuration is required to utilize this enhancement. Corresponding changes in BASE24 (SECUTILS, etc.) are required to fully implement this enhancement.

DEP (Banksys) host security module support

Modifications have been made to the values allowed in the Security Device Configuration UI (and associated data source). A new value of DEP has been added to the allowed Security Device Types. When the DEP device type is selected, the Hardware Model will allow values of DS2, DS3, and DS4 (at this time, only the DS3 hardware model is supported).

In addition to the Security Device Configuration changes, all UIs that display cryptographic keys have been modified. When the DEP HSM is configured, Key fields will be displayed as a single 96-byte field. The full DEP Key token is placed in these fields.

EMV Common Core Definitions (CCD) support

The Authentication UI for EMV configuration has been reorganized to allow for the additional parameters required to configure CCD processing. The scheme selection has been changed from radio buttons to a drop-down selection box. A new value for EMV Scheme has been added for CCD.

In addition, a new parameter, Crypto Version Number, has been added and is a required field. Appropriate values are displayed in this new dropdown based on the EMV scheme selected.

Note

The new Crypto Version Number field must be set for any existing EMV records in the database. This is required for TSS processing of EMV2000 and EMV CCD transactions as the Crypto Version Number may not be supplied in the TSS message from BASE24. For current records using the VISA scheme, the value should be set to '0A' for EMV1996 or '0E' for EMV2000. For current records using the MasterCard/Europay scheme, the value is dependent on the session key derivation method that will be used. Values of '12' or '13' should be selected if the desired session key derivation method is EMV2000. Values of '14' or '15' should be used if the desired session key derivation method is CCD.

Failure to initialize the Crypto Version Number in the EMVSD file will result in receiving the following message at the start-up of the IS process and also from the XML process each time a row is retrieved from the "EMV_SECURITY" datasource.

```
06-12-15;09:07:42.359 \ARGUS.$J3XML  ACI.1049.1000      90
Mgr PPD: \ARGUS.$W23CN Generator: P3C^XML
DASRC: Record read from table = EMV_SECURITY
with a version older than the current version defined in the mdbcsv.
[Sub-system: SIDAL Subject: EMV_Security Action:
sis_rm_data_src_impl::data_buf_version_check() Action Status:
WARNING Error - Metadata has been updated: (The Metadata for this
table has been updated to a newer version. The underlying physical
data layout has been converted to the new format.)]
```

For Offline PIN Change/Unblock support, the Destination PIN Block type drop-down has been moved. Default values are set when EMV Scheme is either VISA or MasterCard/Europay. If the EMV Scheme is 'CCD', the appropriate Destination PIN Block type must be selected by the user.

For more information on valid Crypto Version Number values for EMV, refer to the **BASE24 Transaction Security Services processing guide**.

Re-package TSEC

No additional configuration changes are required for this enhancement.

Configuration for 07.4 enhancements

Eracom Mark II EMV CCD support

An EMV Security record must be configured via the UI. The EMV Scheme and Cryptographic Version Number fields must be a valid EMV CCD combination.

Eracom Mark II EMV uplift

An EMV Security record must be configured via the UI. The EMV Scheme and Cryptographic Version Number fields must be a valid EMV 2000 combination.

Eracom Mark II Keys on CSEC uplift

A user has the option of storing PIN Exchange Keys and MAC Exchange Keys (i.e., Master Keys) in the HSM or in the BASE24-eps database.

If the keys are stored in the HSM, the Key Length drop-down menu on the CSEC UI must be set to "Key Index". The UI will require a 5-digit decimal entry. The range check for the 5-digit decimal entry should be ≤ 65535 .

If the exchange keys are stored in the BASE24-eps database, the Key Length drop-down menu on the CSEC UI must be set to either "Single" or "Double", depending on the key length.

Eracom proof of endpoint uplift

For BASE24-eps TSS, no additional configuration is required to utilize this enhancement.

Eracom Mark II data encryption/decryption product uplift

The "Data Encryption" tab will be enabled on the Channel Security (CSEC) UI when the HSM device type is Eracom Mark II. The information on this tab must be configured for any device that will support data encryption/decryption.

The "Data Encryption Exchange" and "Data Encryption Key Information" tabs will be enabled on the Interface Security (ISEC) UI when the HSM device type is Eracom Mark II. Data on the "Data Encryption Exchange" tab must be configured for each interface that supports dynamic key exchange of data encryption/decryption working keys. Data on the "Data Encryption Key Information" must be configured for any interface that will support data encryption/decryption.

Large message MACing

For BASE24-eps TSS Thales customers who use large message MACing, 07.4 added support of the generic MAC Generate (M6) command. This will require a configuration change.

If converting to 07.4 or later, the Channel Security MAC Option (kam_typ) needs to be changed from 3DES Cipher Block Chaining (1) to Pseudo 3DES Cipher Block Chaining (2) for all records. A MAC Option of Pseudo 3DES Cipher Block Chaining (2) indicates X9.19. The Thales MS command defaulted to ANSI X9.19 MACing. The M6 command allows either ANSI X9.9 or X9.19.

SAMA TSS public key uplift

Support will be added to the RKGGEN (KEYGEN) command to allow identification of the new device type. Allowable values will be INTERFACE, INTFC, or 99.

The Certificate Management and Public Key perusal UIs will support a new device type (a.k.a. Implementation). The new device type (INTERFACE) will be used to indicate that the Public Key in question is for use with a generic Interface configuration.

Support has been added for the new INTERFACE device type when processing public keys for requests from the Certificate Management and Public Key perusal UIs. Processing has been added to handle the specific requirements of the SAMA PKI functionality. This includes formatting the Host Public Key in ASN.1 (ASCII Hex) format without a signature or SHA-1 hash and importing SAMA Root Public Key without signature or SHA-1 Hash.Wincor.

AKDS support of NCR ATMs

For BASE24-eps TSS, no additional configuration is required to utilize this enhancement.

AKDS PCI enhancements

A conversion program exists that will convert the Host Public Key Security Primary (hspkpd) and Host Public Key Security Second (hspksd) tables to the new Host_Public_Key_Security (hspkd) table. The data has been normalized and the Host_Public_Key_Security table contains a record for each key type.

AKDS Capacity report

Two new optional environment parameters are now utilized by this report. These parameters are as follows:

CUSTOMER_NAME — Contains the customer name printed in the header of the report. Default value is 'CUSTOMER NAME'.

AKDS_CAP_RPT_PG_LGTH — Contains the number of lines to be printed on each page of the capacity report. Default value is '60'.

EMV CAP support uplift

A Chip Authentication record must be configured via the UI to utilize this enhancement. In addition, an EMV Security record must be configured via the UI.

Metadata external memory tables

The Installation macro ESSETUP creates the XPNET EMT Build process, and it sets the -MDBV start option to a value of -MT for all other XPNET processes (IS, XML, etc.). The -MT option indicates that the process uses the memory table for Metadata access.

When the Metadata EMT Build program is run, the EMT Build process sends a SIS subscription message to each process started with the MT (Memory Table) value instead of the CSV (Comma Separated Value) for the -MDBV start option. Refer to section 5 for additional information on rebuilding these external memory tables.

Configuration for 08.2 enhancements

MetaMan utility

There are no configuration changes required for this enhancement.

Reorganize non-C++ files

There are no configuration changes required for this enhancement.

DALCI enhancements

There are no configuration changes required for this enhancement.

Release and support strategy

There are no configuration changes required for this enhancement.

Enhanced user ID and password encryption

Customers who choose NOT to impact the PASSWORD and PSWDHIST tables are required to specify the following Application Configuration (APPLCNFG) table property to maintain the current SHA-1 support. Note that salted SHA-256 is more secure than the previous SHA-1 support. For existing users, passwords are migrated from SHA-1 to SHA-256 as the user changes his/her password.

APPLCNFG:

CnfgPropIDTx	PrcsID	CnfgPropValTx	DescrTx
DigestCredAlgorithm	*	SHA-1	SHA-1 password encoding

Configuration for 09.2 enhancements

Since there are no enhancements impacting 09.2 TSS, there are no configuration changes.

Configuration for 11.1 enhancements

For the Interac Flash enhancement:

The EMV tab of the Authentication Profile Configuration UI has new valid values for the EMV Scheme and the Cryptogram Version Number. "Interac" is a new valid value for the EMV Scheme, and "85" is a new valid value for the Cryptogram Version Number. These settings should be used when the record is to be used for Interac Flash EMV Transaction Security functions.

ACI desktop and ACI Java Infrastructure (AJI) considerations

The ESWeb uses a startup and maximum memory allocation of 512 MB. The maximum value for the HP NonStop system depends on the location of the QIO segment. If the segment is moved to KSEG2, the maximum value can be as high as 900 MB. Otherwise, the maximum value may stay around 350 MB.

The latest 11.1 AJPF release contains the following seed data impacts:

APPLCNFG:

Auditing entries have been simplified to remove the individual db.audited.* entries and replace them with the following two entries:

db.audited.datasource.name.aji

db.audited.datasource.name.usec

Several APPLCNFG entries have been modified to make them process-specific using the PrcsID of "APFWeb".

The FieldMasking.xml file was moved from the ESWEB config location to a jar location resulting in the value of the xmlmaskmap.filename change from
`${config.home}/FieldMasking.xml` to
`com/aciworldwide/application/pci/masking/formatter/jaxb/FieldMasking.xml`.

Entries have been added for jsf.jaxb.marshaller.pool.init.count and jsf.jaxb.validator.pool.init.count.

USERS:

A new load script, DUESUSER, was added in version 11.1 to set the new field UserActiveTS to the current date for all users that are preloaded during installation. This field will be set automatically when a new user is added using the User Security Profile Configuration UI. The SIS layer will set the UserExpireTS field to 0.

```
set UserActiveTS = NOW DATETIME (yyyymmddhhmmsslllccc)
```

EVENT:

The parenthesis and square bracket characters that appeared in existing event messages in the EVENT and EVENT_DOCUMENT tables have been removed to ensure the characters used adhere to security best practices for character use and validation.

EXTRCNCT:

A new External Connection entry was added for Application Management. This entry was added with a PrcsID of "APFWeb".

Functs:

The DeleteLimit and DeletePreAuth rows were removed and the Descr text was modified for several entries.

LISTENER:

A new row was added for the Event Adapter.

PASSWORD:

The default password for the pre-loaded User Security Profiles for new users is 'newuser01'. The user will be required to change the password according to the system's password rules. For BASE24-eps, the User Security Profile load scripts contain default password assignments, which should result in no impacts for BASE24-eps customers for the AJI default password changes.

4: Security module requirements

Security module requirements for 03.1 enhancements

Dynamic key management

No specific requirements.

Encrypting PIN Pad (EPP) serial number

No specific requirements. Note that the Encrypting PIN Pad serial number functionality is a part of the Automated Key Distribution System that requires the HSM support Public Key Infrastructure. This is currently limited to certain Atalla and Thales HSMs.

Identikey PIN verification

Supported only on Atalla HSMs.

Data encryption and decryption

Supported only on Atalla HSMs.

Eracom Mark II security module

Support for both Asynchronous and Ethernet attached Eracom security modules.

EMV offline PIN change

Supported on Atalla and Thales HSMs.

New key activation support

No specific requirements.

Security module requirements for 03.2 enhancements

Dutch security

Thales HSM with Dutch Specific Proms is required to support this enhancement.

iCVV verification

No specific requirements.

CVV and CVV2 generation

No specific requirements.

Security module requirements for 03.3 enhancements

Enhanced tracing of HSM messages

No specific requirements.

CAVV verification

No specific requirements.

Interac key management

No specific requirements.

Security module requirements for 04.1 enhancements

Derived Unique Key Per Transaction (DUKPT) support

Support provided for Atalla and Thales HSMs.

Check digit validation/generation utility

Only supported for Atalla HSMs at this time.

Derived Unique Key Per Transaction (DUKPT) MAC support

Only supported for Atalla AKB and Thales HSMs at this time.

EMV Derivation Key Index support

No specific requirements.

Interac changes for BASE24 KEYF

No specific requirements. However, Interac support is currently only provided with Atalla type HSMs.

ISO Format 1 PIN block support

Only supported for Thales type HSMs.

Security module requirements for 04.4 enhancements

AMEX Card Security Code (CSC) validation support

Only supported on Atalla (non-AKB and AKB) and Thales HSMs.

Expiration date in card verify message

No specific requirements.

Encryption type in key generate response

No specific requirements.

Encrypted decimalization table support

Only supported on Atalla (non-AKB and AKB) and Thales HSMs.

Disable public/private key generation (AKDS only)

No specific requirements.

(Note that AKDS is only supported on Atalla AKB and Thales HSMs).

EMV2000 support

Only supported on Atalla AKB and Thales HSMS.

Due to the need to support binary data, support for Async connected Thales devices is not supported.

Minimum firmware versions are required for both Atalla and Thales HSMS. The minimum versions are as follows:

Atalla (all models) 2.62

Thales 7000 7.0a (firmware version 0352.E001) or later.

Thales 8000 2.0a (firmware version 1053.0801) or later.

Eracom AMB hardware security module support

Supported by Eracom AMB HSM only.

MFK/LMK conversion utility for public keys (AKDS only)

Supported by Atalla AKB and Thales HSMS only.

Delivery of source code documentation

No specific requirements.

Delivery of TSS in linkable object form

No specific requirements.

Support for Format 2 files (HP NonStop only)

No specific requirements.

NSJ4 support

No specific requirements.

Security module requirements for 05.2 enhancements

CSEC memory enhancement

No specific requirements.

CSEC Refresh Process Control command

No specific requirements.

Additional security device configuration parameters

If the Security Device Configuration PROTOCOL/INTERFACE parameter is set to TCP/IP or TCP/UDP (RAM or URAM), the MSGMODIFY parameter in the RAM/URAM process configuration must be set to 0. This is because when the security device configuration for the Transaction Security component is set to TCP/IP or TCP/UDP (RAM or URAM), the Transaction Security component no longer adds the STX/ETX to the HSM commands. When RAM/URAM is configured with MSGMODIFY set to 1, it expects the HSM commands to be bracketed by STX/ETX and removes them.

Note: This requirement applies only to Thales HSM environments.

HSM automatic reconnect

No specific requirements.

Security module requirements for 05.4 enhancements

Wincor AKDS support

This enhancement is supported by Atalla AKB and Thales HSMs.

AKDS Capacity report

No specific requirements.

Security module requirements for 06.2 enhancements

APACS MACing support

This enhancement is supported on Thales HSMs only.

X9.19 MACing support

This enhancement provides X9.19 (Pseudo Triple DES) MACing support for the Thales and Eracom (Mark II and AMB) HSMs. Pseudo Triple DES MACing was previously only supported on Atalla HSMs.

VISA DUKPT MACing support

This enhancement provides support for the VISA DUKPT MACing method. Support is provided on the Atalla HSM modules running firmware version 2.82 and later.

Security module requirements for 06.4 enhancements

Contactless chip and magnetic card (dCVV and CVC3) support

Dynamic Card verification has been implemented for Atalla AKB and Thales HSMs only.

A minimum firmware version of 3.11 and above is required for Atalla HSMs.

A minimum firmware version of 2.30 (reference number 1053.0831) and above is required for Thales HSM8000 devices.

A minimum firmware version of 4.0 and 8.0 is required for Thales RG7xxx devices.

CVV1 and CVV2 combined verification support

No specific requirements.

DEP (Banksys) host security module support

Support for Banksys DEP HSM using DS3 message formats and FINTXS firmware.

EMV Common Core Definitions (CCD) support

Support has been implemented for Atalla AKB and Thales HSMs at this time.

Support for Eracom Mark II devices will be added in a future release.

Re-package TSEC

No specific requirements.

Security module requirements for 07.4 enhancements

Eracom Mark II EMV CCD support

Support for Eracom Mark II devices has been implemented.

Eracom Mark II EMV uplift

Support for Eracom Mark II devices has been implemented.

Eracom Mark II keys on CSEC uplift

Support for Eracom Mark II devices has been implemented.

Eracom proof of endpoint uplift

Support for Eracom Mark II devices has been implemented.

Eracom Mark II data encryption/decryption product uplift

Support for Eracom Mark II devices has been implemented.

Large message MACing

Atalla HSM

- Support will be provided for large message MACing for both Variant and AKB HSMs.
- The EMV Offline PIN Change command (350) does not support data chaining. Maximum script data length (including encrypted PIN data) is 3600 bytes.
- The EMV Script MAC Generate command (352) supports data chaining. Support has been implemented for large script data.
- Combined DUKPT PIN Translate/MAC Verify (346) and DUKPT PIN Translate/MAC Generate (347) commands do not support continuation logic. Furthermore, these commands require that the MAC data be in ASCII Hexadecimal format with a maximum field length of 4096. This limits these commands to a real data length of 2048 bytes. If a larger amount of data to be MAC'ed is received, multiple calls to the HSM will be required (a DUKPT PIN Translate (31#7) and two or more DUKPT MAC Verify (348) or DUKPT MAC Generate (386)).
- The Generate MAC Encrypt/Translate Data command (59) does not support continuation logic and is limited to 240 bytes of ASCII data. Large message support has not been implemented for this command.

Bull BNTng HSM

- The Generic MAC Computation (F8), Generic MAC Verification (F9), and EMV MAC Computation (E4) commands are limited to 2048 bytes of data and do not support chaining. Large message MACing has not been implemented for this HSM.

DEP HSM

- EMV Script MAC Generate command (02260D00) does not support data chaining. Script data size is limited to 2048 bytes. Support for large messages has been implemented for this function for the DEP HSM.
- The DUKPT MAC Verify and DUKPT MAC Generate commands currently support a maximum of 2048 bytes of data using the T_STD_TEXT_IN data type. Changes have been made to support the T_STD_TEXT_IN_EXTENDED data type to increase the maximum size to 7.5 Kb as well as support for data chaining.

Eracom HSMs

- Both the Mark II and AMB HSMs currently support very large data sizes in the standard MAC commands (> 255 Mb). However, the overall message length for these devices is dependent on the communication protocol in use (Async = 4096, Ethernet = 1496, TCP/IP = 2048); therefore, modifications to implement data chaining were required.
- The HSM commands used to generate EMV Script MACs (2007 and 2013) do not support data chaining and, therefore, script data will be limited to the device's buffer size (based on communication protocol configured).

Thales HSM

- Current documentation does not indicate that the commands required for supporting full Triple DES (3DES) MACing (MK and MM) support data chaining. These commands are limited to 400 bytes (in Async communications mode) or 800 bytes (binary communications mode) of data.
- Current logic for Single DES MACing only utilizes the MK and MM commands. This has been modified by replacing the use of MK and MM with the MS command to allow data chaining. The MK and MM commands will only be utilized when full Triple DES MACing is configured.
- APACS MACing commands (RM, RS, RI, RK, and RQ) do not support data chaining. Total data size is limited to the HSMs configured buffer size. Support for larger messages has not been implemented as part of this enhancement.
- DUKPT MAC Generate/Verify (GW) command does not support data chaining. Total data size is limited to the HSMs configured buffer size. Support for larger messages has not been implemented as part of this enhancement.
- EMV Script MAC Generate and EMV Offline PIN Change (KU/KW) commands do not allow chaining of the script data being MAC'ed. Total script length is limited to the I/O buffer size of the device.

Support will be added for the generic MAC Generate (M6) and MAC Verify (M8) commands which support chaining.

SAMA TSS public key uplift

Support has been implemented for Atalla AKB, Banksys DEP, and Thales HSMs.

Wincor AKDS support of NCR ATMs

Support has been implemented for Atalla AKB, Banksys DEP, and Thales HSMS.

AKDS PCI enhancements

Support has been implemented for Atalla AKB, Banksys DEP, and Thales HSMS.

AKDS Capacity report

No specific requirements.

EMV CAP support uplift

Support has been implemented for Thales HSMS only.

Metadata external memory tables

No specific requirements.

Security module requirements for 08.2 enhancements

MetaMan utility

No specific requirements.

Reorganize non-C++ files

No specific requirements.

DALCI enhancements

No specific requirements.

Release and support strategy

No specific requirements.

Enhanced user ID and password encryption

No specific requirements.

Security module requirements for 09.2 enhancements

Since there are no enhancements impacting 09.2 TSS, there are no specific security requirements.

Security module requirements for 11.1 enhancements

Interac EMV Flash

Support has been implemented for Atalla AKB.

NCR enhanced Remote Key Loading (RKL) for BASE24 TSS

Support has been implemented for Atalla AKB and Thales HSMS.

5: Other considerations

Additional database updates

Throughout all of the data source updates where the fm_ts is being changed, the examples show that auditing needs to be turned off for the file being updated. This means that the file being updated cannot be open (in other words, the system has to be down). Originally, this was done because DALCI would attempt to perform all updates under a single TMF transaction and larger files would end up failing the update. There is an additional parameter in the UPDATE command called 'NOAUDIT' that forces DALCI to perform Begin/Commits for each row updated. The advantage of this is that the updates can be performed while the system is active. This DALCI enhancement has been available since 04.4 (inclusive).

Using the 'NOAUDIT' option on the DALCI UPDATE command would be especially useful if you have already converted all of your old TSS release data files to the new TSS release, brought your system back up, and start seeing the "version older than the current version defined" warnings. You would not need to bring your system back down to update the particular data source's fm_ts field in order to get rid of the version warnings.

If migrating to version 04.4 or later and you need to update the fm_ts field, the system does not need to be stopped, the FUP ALTERs are not required, and the NOAUDIT parameter should be included in the UPDATE command line.

Example UPDATE of the CSECD data source using NOAUDIT for version 04.4 or later:

```
$BARD T044AKB 365> dalci $<vol>.<subvol> -ai TSS
```

```
DAL Conversational Interface - Version 1.08a
```

```
1: update channel_security fm_ts = 071001 DATE (YYMMDD) noaudit;
```

```
Update all rows in: CHANNEL_SECURITY (y/n): y
```

```
Update of CHANNEL_SECURITY successful - 10027 rows updated.
```

```
2: exit
```

This line is no longer needed before the update if using DALCI UPDATE NOAUDIT:

```
$BARD T044AKB 365> fup alter tsst3data.csecd,no audit
```

This line is no longer needed after the update if using DALCI UPDATE NOAUDIT:

```
$BARD T044AKB 365> fup alter tsst3data.csecd,audit
```

Environment data source

During testing of the 04.1 version it was determined that changes to some of the time processing routines coupled with invalid data in the TSS database would cause errors in the XML Server process. Specifically, invalid file maintenance timestamps (null values) in the

Environment data source cause an abend of the XML server process when attempting to retrieve environment variables from the ACI desktop user interface.

To address this issue, the Environment data source should be updated using DALCI to ensure that no records contain invalid values for the file maintenance timestamp field (user input shown in **bold**):

```
$BARD T041AKB 365> dalci $<vol>.<subvol> -ai TSS
DAL Conversational Interface - Version 1.08a

DALCI> update environment fm_ts = 071001 date (yymmdd) where fm_ts == 0;
DALCI> EXIT
```

The above DALCI command changes the file maintenance timestamp to the equivalent of April 1, 2004 for all records that currently contain a null value in this field.

The above command can be performed at any time including while the online system is active and running.

The above command is only required when migrating a pre-04.1 Environment data source (ENVMTD) to a 04.1 or later version of software. At this time, we do not recommend FUP DUPing this file to your new environment.

Section 6 will instruct you to manually migrate your existing parameters in the ENVMTD into the new environment.

BASE24 KEYF data source

The BASE24 KEYF conversion program at one time did not convert the Interac Key Management fields to valid default values. This will cause a problem when updating KEYF records via the KEYF UI due to invalid field values. The update error may occur even if you do not use the Interac package.

To correct this, the BASE24 KEYF data source should be updated using DALCI to ensure that no records contain invalid values for the Interac Key Management fields (user input shown in **bold**):

```
tacl> dalci $<vol>.<subvol> -ai TSS
DAL Conversational Interface - Version 1.08a

DALCI> update base24_keyf b24_kek_seq_num_out = 0,
                        b24_kek_seq_num_in = 0,
                        b24_out_kme_cur_indx = 1,
                        b24_in_kme_cur_indx = 1;

DALCI> EXIT
```

If you have KEYF files for other logical networks you will need to run the DALCI update command for each KEYF file by appending the logical network id (base24_keyf<lnet>).

You will be instructed when and how to perform the above in section 7.

Enhanced support of versioned tables

Channel security, EMV security, and performance

During performance testing of the 04.4 release it was determined that a change to database versioning processing that was implemented in the SIS layer of code could have an adverse effect on the overall performance when disk I/O is involved. This is due to the additional processing that is required to map a previous version row to the current version row before the data is returned to the business-layer code. (Note that this is not a mandatory step, but for best performance, ACI recommends this step be executed.)

For TSS version 04.4 there are two tables that this modification impacts. The Channel Security data source (CSECD) and EMV Security data source (EMVSD) both have a current version level of 3.0. If rows having a version prior to this (i.e., version 2.0 or version 1.0 rows) are accessed during the online processing, the additional overhead is incurred.

Since the EMVSD is not used directly in online processing (the EMVSOD External Memory Table file built from the EMVSD is used instead) there is no need to explicitly upgrade the rows in this data source (rows are automatically upgraded to the current version level any time an update of a record is performed using the UI).

The Channel Security records, however, can be accessed directly from disk and therefore older row versions can impact online processing. This is particularly true for MAC and data encryption functions that always access disk. As with the EMVSD, records that are updated are automatically upgraded to the current version level. This occurs for updates that occur as a result of file maintenance using the UI or when a record is updated as a result of a key change.

To avoid the additional CPU overhead required to dynamically convert older version records to current version, it is recommended that a global update to all records be performed using DALCI after installation of 04.4 or later software has been completed (including migration of existing data sources to the new installed location(s)). The following command can be used to perform this update (user input shown in **bold**):

```
$BARD T041AKB 365> fup alter tsst3data.csecd,no audit
$BARD T041AKB 365> dalci $<vol>.<subvol> -ai TSS
DAL Conversational Interface - Version 1.08a

1: update channel_security fm_ts = 071001 DATE (YYMMDD);
Update all rows in: CHANNEL_SECURITY (y/n): y
Update of CHANNEL_SECURITY successful - 10027 rows updated.
2: exit
$BARD T041AKB 365> fup alter tsst3data.csecd,audit
```

In the above example, the fm_ts field is used as a common field that can be updated in all records to force the upgrade in the record version information. This field was selected as it provides a common field that can be modified without affecting transaction processing. It is also possible that another field can be used providing that the value is common across all records in the data source.

Note that while the update command can be performed while the system is active, it is recommended that it be performed during a low volume period to avoid data contention issues.

While the EMVSD is not used directly in online processing, any time an EMVSD record is accessed which is not of the current version, you will see the following EMS warning:

```
07-09-25;07:06:31.247 \ACI.$ACI          ACI.1049.1000          90
Mgr PPD: \ACI.$ACI1 Generator: DALCI
DASRC: Record read from table = EMV_SECURITY
with a version older than the current version defined in the mdbcsv.
[Sub-system: SIDAL Subject: EMV_Security Action:
sis_rm_data_src_impl::data_buf_version_check() Action Status:
WARNING Error - Metadata has been updated: (The Metadata for this
table has been updated to a newer version. The underlying physical
data layout has been converted to the new format.)]
```

This is only a warning message and processing continues. However, if you do not want to receive this message every time an EMVSD record is accessed, you can perform the following DALCI global update. User input is shown in **bold** below:

```
$BARD T064AKB 365> fup alter tsst3data.emvsvd,no audit
$BARD T064AKB 365> dalci $<vol>.<subvol> -ai TSS
DAL Conversational Interface - Version 1.08a
```

```
1: update emv_security fm_ts = 071001 DATE (YYMMDD);
Update all rows in: EMV_SECURITY (y/n): y
Update of EMV_SECURITY successful - 10 rows updated.
2: exit
$BARD T064AKB 365> fup alter tsst3data.emvsvd,audit
```

In the above example, the fm_ts field is used as a common field that can be updated in all records to force the upgrade in the record version information. This field was selected as it provides a common field that can be modified without affecting transaction processing. Upon completion of the global DALCI update, your EMVSD records will be updated to the current version.

Diebold Number Table and Identikey conversion issue

*** Note ***

The Thales version of the DNTD cannot be converted using the following steps. The key information is in a "scrambled" form and cannot be manipulated by DALCI. The only options for the Thales DNT implementation is to either reconvert the original data (from the KEYA) or manually enter the data via the (new version) UI from scratch.

During internal QA testing of the 04.4 version of TSS, an issue was discovered related to the BASE24 conversion programs used to convert the KEYA file. This issue affects the Diebold Number Table data source (DNTD) and Identikey data source (IDNT). (Note that this step

would apply to customers going from an earlier version of TSS prior to 04.4, to version 04.4 and later.)

The original conversion program (TSSKEYA) did not properly populate the version field within the records in these data sources. As a result, with the enhanced version support enhancement, the converted records are unusable by the 04.4 version of TSS.

To prevent this problem, the following workarounds are available:

- Reconvert the KEYA file using an updated version of the TSSKEYA conversion utility provided by ACI. Depending on availability of the original KEYA and when the original conversion was performed, this might not be viable.
- Prior to installing TSS version 04.4 or later, using the 04.1 (or prior) version of the TSS XML Server, retrieve each DNT or IDNT record in the UI and perform an update of each record. This sets the version information in each record appropriately and the upgrade to version 04.4 can then be performed normally. (Note that this does not work for version 03.2 or earlier.)
- If you are running v03.2 or earlier, you will need to run a conversion for the DNT and IDNT table.

You will be instructed when and how to perform the above in section 7.

Channel Public Key Security

Four new fields were added for future enhancements to the Channel Public Key Security (CHNPKD) data source in release 06.4, creating version 2 of the CHNPKD.

Additional fields were added for PCI compliance in release 07.4, creating version 3 of the CHNPKD.

If your CHNPKD contains version 1 records and you are using an 06.4 or later release of TSS, you will get the following warning messages in EMS every time a public key is loaded to an ATM:

```
07-09-25;07:06:31.247 \ACI.$ACI          ACI.1049.1000          90
Mgr PPD: \ACI.$ACI1 Generator: DALCI
DASRC: Record read from table = CHAN_PKEY_SEC
with a version older than the current version defined in the mdbcsv.
[Sub-system: SIDAL Subject: Channel_Public_Key_Security Action:
sis_rm_data_src_impl::data_buf_version_check() Action Status:
WARNING Error - Metadata has been updated: (The Metadata for this
table has been updated to a newer version. The underlying physical
data layout has been converted to the new format.)]
```

This is only a warning message and processing continues. However, if you do not want to receive this message every time a public key is loaded to an ATM, you can perform the following DALCI global update after installation of 06.4 or later software (including migration of existing data sources to the new installed location(s)). User input is shown in **bold** below:

```
$BARD T064AKB 365> fup alter tsst3data.chnpkd,no audit
$BARD T064AKB 365> dalci $<vol>.<subvol> -ai TSS
DAL Conversational Interface - Version 1.08a
```

```
1: update chan_pkey_sec fm_ts = 071001 DATE (YYMMDD);
Update all rows in: CHAN_PKEY_SEC (y/n): y
Update of CHAN_PKEY_SEC successful - 10 rows updated.
2: exit
$BARD T064AKB 365> fup alter tsst3data.chnpkd,audit
```

In the above example, the fm_ts field is used as a common field that can be updated in all records to force the upgrade in the record version information. This field was selected as it provides a common field that can be modified without affecting transaction processing. Upon completion of the global DALCI update, your CHNPKD records will be updated to version 2.

Note that while the update command can be performed while the system is active, it is recommended that it be performed during a low volume period to avoid data contention issues.

Metadata external memory table (07.4 and later)

If modifications are made to the Configuration CSV (CONFCSV) or Metadata CSV (MDBCSV) source files, the associated External Memory Table must be rebuilt using the EMT Build process.

- If the EMT Build process is run with the MDB option, the subscription message notifies the SIS Data Factory that a new MDBEMT external memory table is available, and the process automatically switches to the new table.
- If the EMT Build process is run with the CFG option, the subscription message notifies the SIS Data Factory that a new CONFEMT external memory table is available, and the process automatically switches to the new table.
- If the EMT Build process is run with the ALL option, the subscription message notifies the SIS Data Factory that new CONFEMT and MDBEMT external memory tables are available, and the process automatically switches to the new tables.

Note: GoldenGate software does not use the SIS subscription services. You must always stop and restart GoldenGate processes after updating the CONFCSV or MDBCSV. The ESWeb and TDAL processes do not use the SIS subscription services. You must always stop and restart the ESWeb and TDAL processes after updating the CONFCSV or MDBCSV.

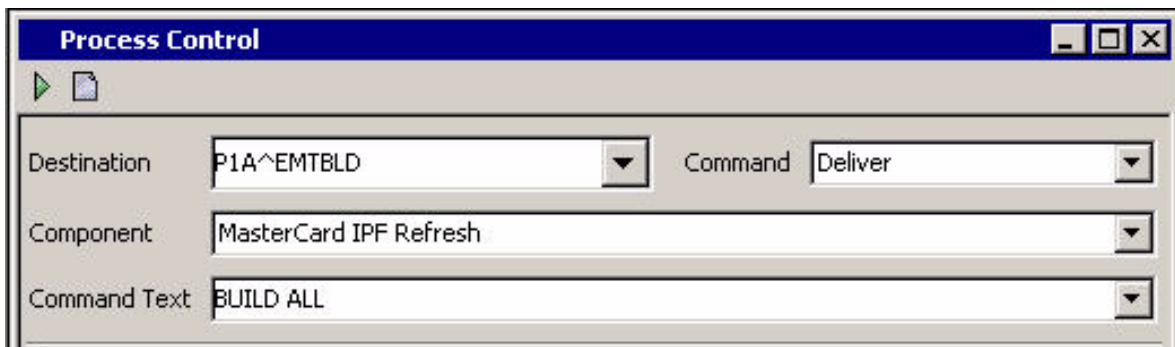
If application coding changes were made in conjunction with the metadata changes, you must also stop and restart TSS processes to pick up any new code and start using the new metadata. You should stop and restart TSS processes on a rolling basis (one at a time) so the system can still process transactions during the transition to new metadata. During the window of time when some processes are using the new version of metadata and some processes are still using the old version of metadata, if a newer version record is read, the SIS DAL layer automatically starts using the newer version instead of returning an error.

To send the Build command from the Process Control window, use the following syntax.

Process Control syntax

Destination:	Metadata EMT Build Symbolic Process Name (e.g., P1A^EMTBLD)
Command:	Deliver
Component:	Select any component from the selectable list. It does not matter which one you select because a component is not used.
Command Text:	BUILD {CFG MDB ALL} For example: BUILD CFG or BUILD MDB or BUILD ALL

The following illustration shows the Build command being sent to the Metadata EMT Build process from the Process Control window.



If you have multiple nodes linked together via XPNET Link processes, the processes in the linked nodes that are started with the MT (Memory Table) value will also receive the broadcast messages.

Refer to the **BASE24-eps MetaMan Utility user guide** for more information on making changes to the ASSIGNS file.

***** Note *****

If migrating to TSS version 07.4 or later and the (default) configuration to utilize Metadata External Memory Tables enhancement is being used (i.e., -MDBV=MT is configured in the STARTOPTIONS parameter of the process configuration), the procedures outlined in "Metadata External Memory Table (07.4 and later)" in section 5 should be performed to reload and warmboot the CONFEMT when modifications are made to the CONFCSV.

Migrating USEC database

***** Note *****

You must convert your database only if you want to keep your existing users. The default users, SysAdmin, SecAdmin, TSSSysAdmin, TSSAdmin, Level1 and Level2 will not be replaced as part of this conversion. These users are enhanced to have accessibility to any new UI records that are added to the new version of TSS. During the conversion, you will receive some "record already exists" messages for these default users. These are acceptable error messages and should be ignored.

This conversion plan assumes that you have a working environment on the release from which you are converting. It also assumes that you have installed a new version of 04.4 or later.

Extracting Enscribe USEC database (04.4 or later) – loading Enscribe USEC database

Extract the old USEC database (<oldprfx> is your old system prefix).

```
volume <oldprfx>cnfg
dalci <oldprfx>cnfg -ai TSS
log to mtrcrc;
select * from actfunc, insertform;
select * from actsess, insertform;
select * from fields, insertform;
select * from fldvals, insertform;
select * from flxfldv1, insertform;
select * from filters, insertform;
select * from functs, insertform;
select * from password, insertform;
select * from pmgpxprm, insertform;
select * from pswdgrp, insertform;
select * from pswdhist, insertform;
select * from perms, insertform;
select * from permdet1, insertform;
select * from permgrp, insertform;
select * from permtype, insertform;
select * from roles, insertform;
select * from rolxpmgp, insertform;
select * from secgrp, insertform;
```

```

select * from srchfld, insertform;
select * from srchxadt, insertform;
select * from taskflds, insertform;
select * from tasks, insertform;
select * from ticket, insertform;
select * from users, insertform;
select * from usrxrole, insertform;
select * from usrstat, insertform;
log stop;
exit;

```

The old USEC data now exists in trace file <oldprfx>data.mtrcrc.

Load the new USEC database (<newprfx> is your new system prefix).

```

volume <newprfx>cnfg
dalci <newprfx>cnfg -ai TSS
obey <oldprfx>data.mtrcrc;
exit;

```

Note: You will receive some "record already exists" errors. These errors can be ignored as all records from the old system should now exist in the new system, except for the default users (SysAdmin, SecAdmin, TSSSysAdmin, TSSAdmin, Level1 and Level2).

Extracting any version SQL USEC database (or 0.41 and older Enscribe database) – loading Enscribe USEC database

HP NonStop Database Extract

1. Determine the starting templates.

Note: If you are on a release that is prior to 03.3, use the EXTRACT32.properties and LOAD32.cfg files instead of the EXTRACT.properties and LOAD.cfg files referenced in the following steps. You must also rename the xxx32 files to xxx if you are on a release prior to 03.3.

2. Examine the USEC.cfg file in your old environment (the environment you want to extract from 03.2, 03.3, etc.)
3. Update the EXTRACT.cfg file.
 - a. In the [COMMON] section, replace the entries that are blank with the corresponding entries out of your old environment USEC.cfg [COMMON] section.
 - b. In the [DBMANAGER] section, replace the entries that are blank with the corresponding entries out of your old environment USEC.cfg [DBMANAGER] section.
 - c. In the [EXTRACT] section, replace {ESHOME} reference with the root of your old environment and verify the "PropertyFile=" points to the LOAD properties file, which will be created for you during the RUNXTRA procedure below. This file will be created on your old environment location DBEXTRACT, which will also be created for you in a later step.
 - d. Ignore the LOAD section in this file.

4. Update the RUNXTRA file.
 - a. Update all references of {ESHOME} to point to the root of your old environment. When you are done, the "export EXTRACTFILE=" line should point to a fully qualified EXTRACT.cfg file, and the {ESHOME}/ThirdParty/SetCommon line should point to the SetCommon file on your old environment.
 - b. Examine the RunUSECTacl file in the old database.
 - c. Update the export JDBCdriver= to match the definition from the RunUSECTacl file on your old system.
 - d. Update the "export USECDB=" value. If you are using SQLMP/SQLMX for the USEC database, then you also must replace {Location of SQL tables if SQLMP/SQLMX} with the volume and subvolume of the USEC table location of your old environment. If you are not using SQLMP/SQLMX, delete all text to the right of the equal sign (including quotes).
 - e. Update the -name=/G/DBLD value to be the name of a PPD that is compatible with your naming schemes.
5. Create a DBEXTRACT directory on your old environment directly under the {ESHOME} location.
 - a. cd {ESHOME}.
 - b. mkdir DBEXTRACT.
6. Transfer the Extract files from your PC to the HP NonStop.
 - a. Transfer the RUNXTRA, EXTRACT.cfg, and EXTRACT.properties files in ASCII to the DBEXTRACT directory that you just created above.
 - b. Transfer the cef.jar and UIFramework.jar, in binary, to the DBEXTRACT directory that you just created above.
 - c. Change the mode on the RUNXTRA file so that it is executable on your old environment (chmod 777 RUNXTRA).
7. Run the extract object.
 - a. cd {ESHOME}/DBEXTRACT.
 - b. ./RUNXTRA (This is a dot followed by a forward slash at the command line.)
 - c. You can ignore the following errors messages:


```
ACI ES Unable to log events during impl creation for SSID = 1049, Event Number = 20
[Sub-system: SIDAL Subject: FILE_PART_LIST Action: sis_db_factory_impl::data_src_create()]
Action Status: FAILED Error dsrc_asgn_not_found: (FILE_PART_LIST)] ACI ES
Unable to log the following event during impl creation: SSID = 1008, Event Number = 20,
Component ID = DASRC
```

And depending on which version, you may receive the following which can also be ignored:

```
Database extract in progress...

LogScrubber: Unable to locate scrub tag configuration file(s): LogScrubber.cfg

Prepared Statement caching will be enabled
```
8. If the process ran correctly, you should now have a LOAD.properties file in your {ESHOME}/DBEXTRACT directory in your old environment.

NT Database Extract

1. Determine the starting templates.

Note: If you are using an MS SQL database on the Windows NT platform, you must use the EXTRACT_MS.cfg and EXTRACT32_MS.properties instead of the EXTRACT.cfg and EXTRACT.properties referenced in the following steps.

2. Update the EXTRACT_MS.cfg file.
 - a. In the [COMMON] section, replace the entries that are blank with the corresponding entries out of your old environment USEC.cfg [COMMON] section.
 - b. In the [DBMANAGER] section, replace the entries that are blank with the corresponding entries out of your old environment USEC.cfg [DBMANAGER] section.
 - c. Update the {ESHOME} reference to point to the root of your old environment. After it has been altered, the PropertyFile= under [EXTRACT] should point to the LOAD.properties file.
 - d. You can ignore the LOAD section in this file RunExtract.bat.
3. Update the RunExtract.bat file if the database is MS SQL on a Windows NT machine.
 - a. Update the {ESHOME} reference to point to your old installation directory. Example: C:_ACI_UI.
4. Create a DBEXTRACT directory on your old environment directly under the {ESHOME} location.
 - a. Transfer the RunExtract.bat, EXTRACT_MS.cfg and EXTRACT32_MS.properties to the DBEXTRACT directory that you created.
 - b. Transfer the cef.jar and UIFramework.jar to the DBEXTRACT directory that you created.
5. Run the extract object.
 - a. Run the RunExtract.bat file to start the extraction process. The resulting LOAD.properties file should be transferred to the HP NonStop {ESHOME}/DBEXTRACT directory created in step 4.
6. If the process ran correctly, you should now have a LOAD.properties file in your {ESHOME}\DBEXTRACT directory.

HP NonStop Database Load

1. Create a DBLoad.bat file on your PC containing the following:

```
echo off
set JREBin= C:\jdk1.5.0_05\jre

rem ***** do not change anything below this line *****

set props=%1%
if not defined props set props=LOAD.cfg

set CLASSPATH=uif_srv.jar;uif_sh.jar
```

```

set CLASSPATH=%CLASSPATH%;jdal.jar
set CONFIGS=-DConfig=%props%

echo %JREBin%\java -classpath %CLASSPATH% %CONFIGS% cef.shared.DBLoad
%JREBin%\java -classpath %CLASSPATH% %CONFIGS% cef.shared.DBLoad

set props=
set root=
set prod=
set jdbc=
set CONFIGS=
set CLASSPATH=
set JREBin=

pause

```

2. Create a LOAD.cfg file on your PC containing the following:

```

[Common]
LogPath=.\Logs
LogFileName=ES_DB_USEC
AllowErrors=true
MaxLogKilobytes=900
Policy=.\uif.policy
PolicyAutoUpdate=false

[DBMANAGER]

drivers=com.aciworldwide.jdbcdriver.jdal.JDALDriver

ESF.url=jdbc:JDAL://{Replace with NSK host}:{Replace with port BASE+6}
ESF.user=pserve
ESF.password=
ESF.initconns=5
ESF.maxconns=10
ESF.logintimeout=200

[EXTRACT]
DBName=ESF
PropertyFile=.\ESF.properties
MetaDataOnly=false

[LOAD]
OwnerIdConvert=false
DBName=ESF
PropertyFile=.\LOAD.properties
LoadType=BOTH

```

3. Place the DBLoad.bat and LOAD.cfg files created above into a PC directory.
4. Copy the LOAD.properties file that was generated from the extract into the same directory as the above files.

5. Copy the uif_srv.jar, uif_sh.jar, and jdal.jar into this directory. (These should be in the shipment area.)
6. Edit LOAD.cfg to point to database. ESF.url=jdbc:JDAL://{Replace with host}:{Replace with port BASE+6}
7. Edit DBLoad.bat to point to a valid JRE.
set JREBin= C:\jdk1.5.0_05\jre
8. Run the program with "DBLoad LOAD.cfg".

Migrating TSS database

After successful completion of the standard TSS installation procedures, you will now need to migrate your old database to the new database based on your version of TSS.

- **Do not move** the following TSS data sources (typically located on the TSSTDATA subvolume)
 - ☐ ACTFUNC1¹
 - ☐ ACTSESS2
 - ☐ APPLDSRC
 - ☐ APPLD
 - ☐ AUDDPK1¹
 - ☐ AUDDTL1¹
 - ☐ AUDTBL¹
 - ☐ CMSGD
 - ☐ EVDOCD
 - ☐ EVENTD
 - ☐ EVTLOG
 - ☐ ENVMTD

*** If you have added/modified environment parameters to meet your site's needs, reapply these changes manually to the Environment data source.

 - ☐ EXTCND
 - ☐ FPLSOD
 - ☐ FPLSTD
 - ☐ KEYF
 - ☐ MSGCTXT
 - ☐ OPDOCD
 - ☐ OPDOCDO

¹ These files existed in releases 03.3 and 04.1 only.

- ❑ PROCD
- ❑ TRNSD
- ❑ TRNSOD
- ❑ UALOGD
- ❑ UALOGD0
- ❑ USRXFILT
- ❑ VERSD
- **Do not move** the following security files as their migration is covered in section 7.
 - ❑ CSECD
 - ❑ DNTD
 - ❑ DNTOD
 - ❑ HSPKD
 - ❑ HSPKPD
 - ❑ HSPKSD
 - ❑ IDNTD
 - ❑ IDNTOD
 - ❑ ISECD
 - ❑ ISECOD
 - ❑ SECD
 - ❑ SECD0

The files listed below now need to be copied from your old database to your new database.

The system must be stopped. Perform a FILEINFO on the DATA subvolume to get a listing of AUDITED files. Move the following TSS data sources (typically located on the TSSTDATA subvolume).

Note: You could exclude from the list below any old data sources that do not have data in them, i.e., 0 EOF/empty files.

ASAFCD	ASAFD	ASAFOD	BNTHD	BNTHOD	BNTID
CAD	CAOD	CHNPKD	CRDVD	CRDVOD	DALTBL
DCRDOD	DCRDVD	DOLTPD	EMVSD	EMVSOD	EPNUMD
EXLOGD	HSMCNF	IDESD	IDESOD	LSTND	MDSDEST*
NCRD	NCROD	TRCCFD	TRCFLD	USAFCD	USAFD
USAFOD	VPVVD	VPVVOD			

* If PPD names for your processes have not changed, the existing previous version MDSDEST may be copied to the new data location. If the PPD names have changed, edit the MDSDESTS file (produced by SETUPES or ESSETUP) to update the security device names, and obey MDSDESTC to reload the MDSDEST.

The text on the following page can be used as a template command file to perform the copies of the above files. Note that depending on your old version, some of these files might not exist, and the default files created by the install process can be used.

```
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.ASAFCD      , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.ASAFD        , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.ASAFOD       , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.BNTHD        , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.BNTHOD       , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.BNTID        , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.CAD          , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.CAOD         , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.CHNPKD       , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.CRDVD        , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.CRDVOD       , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.DALTBL       , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.DCRDOD       , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.DCRDVD       , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.DOLTPD       , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.EMVSD        , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.EMVSOD       , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.EPNUMD       , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.EXLOGD       , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.IDESD        , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.IDESOD       , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.LSTND        , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.MDSDEST      , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.NCRD         , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.NCROD        , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.TRCCFD       , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.TRCFLD       , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.USAFCD       , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.USAFD        , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.USAFOD       , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.VPVVD       , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
fup dup $<OLD_VOLUME>.<OLD_SUBVOL>.VPVVOD      , $<NEW_VOLUME>.<NEW_SUBVOL>.*,purge,sourcedate
```

Some of the above OLTP files may not exist on your system if you do not use that functionality.

Note: For versions prior to v05.2, there will be no trc* files.

Using the FILEINFO output from the previous page, fup alter those files that are audited.
The syntax is:

```
FUP ALTER <filename>, AUDIT
```

Now that you have FUP DUPed a majority of the old data sources to the new environment, proceed with section 7.

B. Database conversions

Channel Security data source conversion

***** Note *****

Prior to performing the steps outlined in this section, current backups of the existing CSECD file should be performed.

***** Note *****

These procedures should only be performed when the system is not active.

If migrating from v02.4:

- If you have increased the size of your old CSECD, alter the new CSECD in the same manner.

```
FUP COPY <old_vol_subvol>.CSECD, <new_vol_subvol>.CSECD, REOUT 1433,  
PAD 0
```

- Proceed with step 1.

If migrating from v03.1 or v03.2:

- FUP DUP the old CSECD to the new data location.
- Proceed with step 1.

If migrating from v03.3 or v04.1:

- FUP DUP the old CSECD to the new data location.
- Skip to step 2.

If migrating from v04.4, v05.2, v05.4, v06.2, v06.4, v07.4, or v08.2:

- FUP DUP the old CSECD to the new data location.
- Skip to step 2b.

If migrating from v09.2:

- FUP DUP the old CSECD to the new data location.
- Skip to step 2c.

1. Initialize the new fields in the CSECD to default values using the DALCI executable included with your new version.
 - a. If migrating from v02.4, enter the following command (the U031CSEC file can be created using information found in section 8):

```
DALCI> OBEY U031CSEC;
```

- b. If migrating from v02.4, v03.1, or v03.2, enter the following command (the U033CSEC file can be created using information found in section 8):

```
DALCI> OBEY U033CSEC;
```

- 2. Initialize the new fields in the CSECD to default values using the DALCI executable included with your new version.

- a. If migrating from v03.1, v03.2, v03.3, or v04.1, enter the following command (the U044CSEC file can be created using information found in section 8):

```
DALCI> OBEY U044CSEC;
```

- b. If migrating from v08.2 or earlier, also enter the following command (the U092CSEC file can be created using information found in section 8):

```
DALCI> OBEY U092CSEC;
```

- c. If migrating from v09.2 or earlier, also enter the following command (the U111CSEC file can be created using information found in section 8):

```
DALCI> OBEY U111CSEC;
```

3. FUP ALTER CSECD, AUDIT

The CSECD is now ready for use. Note that to perform data encryption functions, the appropriate key fields must be populated using the BASE24-eps UI. The encryption parameters for each interchange or channel should also be reviewed (using the UI) to insure that the default values are sufficient for your system.

Proceed to "Interchange Security data source conversion."

Interchange Security data source conversion

*** Note ***

Prior to performing the steps outlined in this section, current backups of the existing ISECD file should be performed.

*** Note ***

These procedures should only be performed when the system is not active. Any database changes made (typically as a result of dynamic key changes) after completion of step 1 below will be lost during the conversion process.

If migrating from v03.2 or later:

- FUP DUP the old ISECD to the new data location.
- Skip to step 6c.

The following procedures are only required when migrating from versions of TSS prior to 03.2. Database versioning is used for handling database changes in release 03.3 and forward.

To convert the old ISECD data file to the new format required with this version of TSS, the following steps should be performed:

1. In the old data source area, create a DALCI load script using your old DALCI executable for the old data. This can be accomplished using DALCI and the following commands:

```
DALCI> LOG TO TRC;
DALCI> SELECT * FROM INTERCHANGE_SECURITY INSERTFORM HEX;
DALCI> LOG STOP;
```

2. In most cases, the TRC data source is to a spooler location.

```
"TRC","Stream","1.0","$s.#trc","WOF","",""
```

You will need to do a 'list' of the data to an edit file.

```
TACL> PERUSE
_ J<job #>
_ list E /out tesldata.LDISEC !/ f/1
_ EXIT
```

Now the old data has been extracted into a temporary file. In a subsequent step, you will load these records into the new database.

3. Edit the LDISEC file and change all references of INTERCHANGE_SECURITY to ICHG_SECURITY.
4. Note that the new CONFCSV (Configuration) and MDBCSV (Record Layout) files were automatically placed on the newly installed version of TSS and are used in the following DALCI load step.
5. Using the DALCI executable included with your new version, reload the ISECD using the DALCI script created in the previous step. (Note that this loads the extracted file from the old database into the new database created during the install of the new version of TSS.) This can be accomplished using the following command:

```
DALCI> OBEY LDISEC;
```

6. Initialize the new fields in the ISECD to the default values using the DALCI executable included with your new version.

- a. If migrating from v02.4, enter the following command (the U031ISEC file can be created using information found in section 8):

```
DALCI> OBEY U031ISEC;
```

- b. If migrating from v02.4 or v03.1, also enter the following command (the U032ISEC file can be created using information found in section 8):

```
DALCI> OBEY U032ISEC;
```

- c. If migrating from v03.2, v03.3, v04.1, v04.4, v05.2, v05.4, v06.2, v06.4, v07.4, or v08.2, also enter the following command (the U092ISEC file can be created using information found in section 8):

```
DALCI> OBEY U092ISEC;
```

- d. If migrating from v09.2 or earlier, also enter the following command (the U111ISEC file can be created using information found in section 8):

```
DALCI> OBEY U111ISEC;
```

7. The Interchange Security OLTP (ISECOD) can now be re-loaded either using DALCI or from the Process Control function within the BASE24-eps UI environment.

```
DALCI> LOAD OLTP INTERCHANGE_SECURITY_OLTP;
```

8. FUP ALTER ISECD, AUDIT

The ISECD is now ready for use. Note that to perform data encryption functions, the appropriate key fields must be populated using the BASE24-eps UI. The encryption parameters for each interchange or channel should also be reviewed (using the UI) to insure that the default values are sufficient for your system.

Proceed to "Security Device Configuration data source conversion."

Security Device Configuration data source conversion

*** Note ***

Prior to performing the steps outlined in this section, current backups of the existing SECD file should be performed.

*** Note ***

The name of the security device(s) must match what is in the MDSDEST. If you are going to manually enter the security devices, these devices must also match the names in the MDSDESTS & MDSDEST.

1. Fup dup the 'old' SECD CD to the new data location or re-enter the data through the new BASE24-eps UI environment.
2. You will now need to turn audit on:

```
FUP ALTER SECD CD, AUDIT
```

3. If migrating from v05.2 or later or if you have re-entered the data through the BASE24-eps UI, skip to "Diebold Number Table conversion." The following conversion does not need to be performed.

The steps outlined below apply to conversion of the Security Device Configuration data source when you are migrating from any version prior to 05.2 and you have not re-entered the data through the new BASE24-eps UI environment.

4. The Security Device Conversion utility produces an execution report using the ES Foundation Layer Report Stream function. The output is directed to the MTRCRC data source that is referenced in the Metadata Configuration file, CONFCSV. The following is an excerpt from the HP NonStop CONFCSV file.

```
"MTRCRC", "Stream", "1.0", "$vol.subvol.mtrcrc", "WOF", "", ""
```

5. Before running the following conversion do a fup purgedata on the TSSTDATA.MTRCRC.
6. Run the Security Device Configuration conversion program, provided with this release, against the database, using the following command:

```
TACL> volume TSSTOBJ
```

```
TACL> RUN SDCCNVLO/NAME [<ppd_name>], LIB <xpnet_library_name>/-MDBV=MT -AI=TSS -
STRT=<confcsv_vol>.<confcsv_subvol>
```

Where:

<ppd_name> = Valid HP Nonstop process name (i.e., \$CNV01). If no name is supplied, the system generates a name automatically.

<xpnet_library_name> = Location and name of the XPNET Library file (i.e., \$<VOL>.XNLIBN.XNLIBN, if using XPNET 3.0, or \$<VOL>.XNLIB.XNLIBN, if using XPNET 3.1).

<confcsv_vol>.<confcsv_subvol> = Volume and subvolume location of the (version 05.2 or later) CONFCSV/MDBCSV files. (i.e., \$DATA.TSSTCNFG).

The default location for the MTRCRC file in the CONFCSV file is the same location for all other data sources, so the report output in this example would end up in the \$vol.subvol that was specified by the installation process. To view:

```
TACL> FUP COPY MTRCRC:
```

```
Security Device Configuration Conversion Report - Jan 31, 2005 22:37:22
```

```
**** Record updated - dev_name   = ERACOM1
****                  - dev_ttyp  = E
****                  - dev_subtyp = 2
****                  - hdwr_md1  = 7
****                  - proto_intf = 2
****                  - key_scheme = 0
**** Record updated - dev_name   = ERACOM2
```



```

****                - dev_typ      = E
****                - dev_subtyp    = 2
****                - hdwr_md1     = 7
****                - proto_intf    = 2
****                - key_scheme    = 0
*****
****  Number of Records updated - 2  ****
****  Conversion ran successfully.   ****
*****

```

7. If you have one of the following devices and use the corresponding default protocol, you can proceed to "Diebold Number Table conversion."

ATALLA - always defaults to 3 (boxcar - TCP/IP).

RACAL - defaults to 1 (ASYNC) or 4 (TCP/IP), depending on the device subtype.

ERACOM - defaults to 1 (ASYNC) or 2 (ETHERNET), depending on the device subtype.

If not, you will need to update your newly converted SECD CD file with the correct protocol type number. To do this, you will need to use the new DALCI that was sent and issue the following command:

```

DALCI> UPDATE SEC_DEV_CONFIG proto_intf = <protocol number>;
Where <protocol type number> =
1 = ASYNC
3 = ATALLA with TCP/IP
4 = RACAL with RAM
4 = ERACOM with TCP/IP
7 = UDP (RACL with URAM)

```

The SECD CD is now ready for use.

Proceed to "Diebold Number Table conversion."

Diebold Number Table conversion

*** Note ***

Prior to performing the steps outlined in this section, current backups of the existing DNTD file should be performed.

*** Note ***

These procedures should only be performed when the system is not active. Any database changes made, typically as a result of dynamic key changes, after completion of step 1, 2, or 3 below will be lost during the conversion process.

*** Note ***

If migrating from versions of TSS prior to v04.4 and using Thales, the DNTD cannot be converted using the following steps. The key information is in a 'scrambled' form and cannot

be manipulated by DALCI. The only options for the Thales DNT implementation is to either reconvert the original data (from the KEYA) or manually enter the data via the (new version) UI from scratch, then skip to "Identikey conversion."

If migrating from v04.4 or later and using Thales, fup dup the 'old' DNTD to the new data location and skip to step 4, loading the OLTP.

1. If migrating from v04.4 or later:

FUP DUP the 'old' DNTD to the new data location and skip to Step 4, loading the OLTP.

2. If migrating from v03.3 or v04.1:

Using your old BASE24-eps UI, retrieve each DNTD record in the UI and perform an update of each record. This sets the version information in each record appropriately.

FUP DUP the 'old' DNTD to the new data location and skip to Step 4, loading the OLTP.

3. If migrating from v02.4, v03.1 or v03.2:

The following conversion is required. To convert the old DNTD data file to the new format required with this version of TSS, the following steps should be performed:

- a. In the old data source area, create a DALCI load script using your old DALCI executable for the old data. This can be accomplished using DALCI and the following commands:

```
DALCI> LOG TO TRC;
DALCI> SELECT * FROM DIEBOLD_NUMBER_TABLE INSERTFORM HEX;
DALCI> LOG STOP;
DALCI> EXIT
```

- b. In most cases, the TRC data source is to a spooler location. You will have to 'LIST' the data to an edit file.

```
"TRC","Stream","1.0","$s.#trc","WOF","",""
```

```
TACL> PERUSE
_ J<job #>
_list E /out tesldata.LDDNTD !/ f/1
_EXIT
```

- c. Edit the LDDNTD file and change all references of DIEBOLD_NUMBER_TABLE to DIEBOLD_NUM_TBL.
- d. Note that new CONFCSV (Configuration) and MDBCSV (Record Layout) files were automatically placed on the newly installed version of TSS and are used in the following DALCI load step.
- e. Using the DALCI executable included with your new version, reload the new DNTD using the DALCI script created in a previous step. (Note that this will load the

extracted file from the old database into the new database created during the install of the new version of TSS.) This can be accomplished using the following command:

```
DALCI> OBEY LDDNTD;
```

4. The Diebold Number Table OLTP (DNTOD) can now be re-loaded either using DALCI or from the Process Control function within the BASE24-eps UI environment.

```
DALCI> LOAD OLTP DIEBOLD_NUMBER_TABLE_OLTP;
```

5. You will now need to turn audit on:

```
FUP ALTER DNTD, AUDIT
```

The DNTD is now ready for use.

Proceed to "Identikey conversion."

Identikey conversion

*** Note ***

Prior to performing the steps outlined in this section, current backups of the existing IDNTD file should be performed.

*** Note ***

These procedures should only be performed when the system is not active. Any database changes made, typically as a result of dynamic key changes, after completion of step 1, 2, or 3 below will be lost during the conversion process.

1. If migrating from v04.4 or later:

FUP DUP the 'old' IDNTD to the new data location and skip to Step 4, loading the OLTP.

2. If migrating from v03.3 or v04.1:

Using your old BASE24-eps UI, retrieve each IDNTD record in the UI and perform an update of each record. This sets the version information in each record appropriately.

FUP DUP the 'old' IDNTD to the new data location and skip to Step 4, loading the OLTP.

3. If migrating from v02.4, v03.1 or v03.2:

The following conversion is required. To convert the old IDNTD data file to the new format required with this version of TSS, the following steps should be performed:

- a. In the old data source area, create a DALCI load script using your old DALCI executable for the old data. This can be accomplished using DALCI and the following commands:

```
DALCI> LOG TO TRC;
DALCI> SELECT * FROM IDENTIKEY INSERTFORM HEX;
DALCI> LOG STOP;
```

```
DALCI> EXIT
```

- b. In most cases, the TRC data source is to a spooler location. You will have to 'LIST' the data to an edit file.

```
"TRC","Stream","1.0","$.#trc","WOF","",""
```

```
TACL> PERUSE
_ J<job #>
_ list E /out tesldata.LDIDNTD !/ f/1
_ EXIT
```

- c. Note that new CONFCSV (Configuration) and MDBCSV (Record Layout) files were automatically placed on the newly installed version of TSS and are used in the following DALCI load step.
- d. Using the DALCI executable included with your new version, reload the new IDNTD using the DALCI script created in a previous. (Note that this will load the extracted file from the old database into the new database created during the install of the new version of TSS.) This can be accomplished using the following command:

```
DALCI> OBEY LDIDNTD;
```

4. The Identikey OLTP (IDNTOD) can now be re-loaded either using DALCI or from the Process Control function within the BASE24-eps UI environment.

```
DALCI> LOAD OLTP IDENTIKEY_OLTP;
```

5. You will now need to turn audit on:

```
FUP ALTER IDNTD, AUDIT
```

The IDNTD is now ready for use.

Proceed to "KEYF conversion."

KEYF conversion

*** Note ***

Prior to performing the steps outlined in this section, current backups of the existing KEYF file should be performed.

*** Note ***

If you have KEYF files for other logical networks, you will need to do any needed updates below for each KEYF listed in your CONFCSV.

For example, if these are your CONFCSV KEYF entries:

```
"BASE24_KEYF","BASE24_KEYF","1.0","$ESTS.TS92DATA.keyfd","Enscribe","",""
"BASE24_KEYFDEV1","BASE24_KEYF","1.0","$ESTS.TS92DAT1.keyf","Enscribe","",""
"BASE24_KEYFDEV2","BASE24_KEYF","1.0","$ESTS.TS92DAT2.keyf","Enscribe","",""
```

and your BASE24_KEYF KEYFD file is empty, each of your conversion update steps below will need to be done for each KEYF file:

```
UPDATE BASE24_KEYFDEV1 ... ;
UPDATE BASE24_KEYFDEV2 ... ;
```

If migrating from v06.2 or earlier, the following conversion of the BASE24 KEYF data source is required.

1. Create the DALCI obey file U062KEYF to contain the following DALCI script:

```
UPDATE BASE24_KEYF
b24_msg_key_txn_lmt = 0,
b24_msg_key_timr_lmt = 0,
b24_msg_cnsc_err_lmt = 0,
b24_msg_err_lmt = 0
;
```

If migrating from v03.2, you will need additional lines in your U062KEYF DALCI obey file. After adding these lines, your file should look like this:

```
UPDATE BASE24_KEYF
b24_kek_seq_num_out = 0,
b24_kek_seq_num_in = 0,
b24_out_kme_cur_indx = 1,
b24_in_kme_cur_indx = 1,
b24_msg_key_txn_lmt = 0,
b24_msg_key_timr_lmt = 0,
b24_msg_cnsc_err_lmt = 0,
b24_msg_err_lmt = 0
;
```

If migrating from v06.2 or earlier to v06.4 or later, in addition to either of the applicable UPDATES above, you will also need the following UPDATE in your U062KEYF DALCI obey file:

```
UPDATE BASE24_KEYF
b24_full_encrypt = "0",
b24_encrypt_typ_msg = "0"
;
```

- Using the DALCI executable included with your new version, initialize the above fields in the KEYF to default values by entering the following command:

```
DALCI> OBEY U062KEYF;
```

The KEYF is now ready for use.

Proceed to "Host Public Key data source conversion."

Host Public Key data source conversion

*** Note ***

Prior to performing the steps outlined in this section, current backups of the existing HSPKPD and HSPKSD files should be performed.

- If migrating TO v06.4 or earlier:
FUP DUP the 'old' HSPKPD and 'old' HSPKSD to the new data location or re-enter the data through the new BASE24-eps UI environment, and skip to step 6, Turn audit on.
- If migrating FROM v07.4 or later:
FUP DUP the 'old' HSPKPD to the new data location or re-enter the data through the new BASE24-eps UI environment, and skip to step 6, Turn audit on.
- If migrating FROM any version earlier than v07.4, and migrating TO any version later than v06.4:
FUP DUP the 'old' HSPKPD and 'old' HSPKSD to the new data location or re-enter the data through the new BASE24-eps UI environment.

The steps outlined below apply to conversion of the Host Public Key Security data source when you are migrating from any version prior to 07.4 to any version later than 06.4 and you have not re-entered the data through the new BASE24-eps UI environment.

- The Host Public Key Conversion utility produces an execution report using the ES Foundation Layer Report Stream function. The output is directed to the data source referenced by the PUBLIC_KEY_CONV_RPT assign in the CONFCSV. For example:
"PUBLIC_KEY_CONV_RPT", "Stream", "1.0", "\$S.#PUBKEY", "WOF", "", ""
- Run the Host Public Key Security conversion program, provided with this release, against the database, using the following command:

```
TACL> volume TSSTOBJ
```

```
TACL> RUN HPKCNVLO/NAME [<ppd_name>], LIB <xpnet_library_name>/-MDBV=CSV -  
AI=TSS -STRT=<confcsv_vol_subvol>
```

Where:

<ppd_name> = Valid HP Nonstop process name (i.e., \$CNV01). If no name is supplied, the system generates a name automatically.

<xpnet_library_name> = Location and name of the XPNET Library file (i.e., \$<VOL>.XNLIBN.XNLIBN, if using XPNET 3.0, or \$<VOL>.XNLIB.XNLIBN, if using XPNET 3.1).

<confcsv_vol_subvol> = Volume and subvolume location of the (version 05.2 or later) CONFCSV/MDBCSV files (i.e., \$DATA.TSSTCNFG).

The default location for the PUBLIC_KEY_CONV_RPT file in the CONFCSV file is a spooler location. The following is a sample report:

```
Host Public Key Data Source Conversion Report - Oct 17, 2007 09:36:48
```

```
-----
Unable to obtain validity dates for the record with following
keys :
```

```
-----
Host Name      : DLKHOST1
Device Type    : 1
```

```
RECORD NOT INSERTED !!!
```

```
-----
PUBLIC KEY DATA SOURCE CONVERSION COMPLETED SUCCESSFULLY !!!
```

```
Number of Records inserted : 14
```

```
Host Public Key Primary Data Source Records processed : 6
```

```
Host Public Key Secondary Data Source Records processed : 2
-----
```

6. Turn audit on:

If migrating TO v06.4 or earlier:

```
FUP ALTER HSPKPD, AUDIT
FUP ALTER HSPKSD, AUDIT
```

Otherwise:

```
FUP ALTER HSPKD, AUDIT
```

The HSPKD is now ready for use.

Proceed to "Derivation Key data source conversion."

Derivation Key data source conversion

*** Note ***

Prior to performing the steps outlined in this section, a current backup of the existing DRVKD file should be performed.

1. If migrating to v08.2 or earlier:

FUP DUP the 'old' DRVKD to the new data location or re-enter the data through the new BASE24-eps UI environment, and skip to step 4, loading the OLTP.

2. If migrating to 09.2 or later:

FUP DUP the 'old' DRVKD to the new data location or re-enter the data through the new BASE24-eps UI environment.

Create the DALCI obey file U092DRVK to contain the following DALCI script:

```
UPDATE DERIVATION_KEY
encrypt_des_mde = 1
;
```

3. Using the DALCI executable included with your new version, initialize the above field in the DRVKD to a default value by entering the following command:

```
DALCI> OBEY U092DRVK;
```

4. The Derivation Key OLTP (DRVKOD) can now be re-loaded either using DALCI or from the Process Control function within the BASE24-eps UI environment.

```
DALCI> LOAD OLTP DERIVATION_KEY_OLTP;
```

5. Turn audit on:

```
FUP ALTER DRVKD, AUDIT
```

The DRVKD is now ready for use.

Please review Section 5 for optional database updates which may improve performance or suppress unwanted event messages.

YOUR ENTIRE UPGRADE IS NOW COMPLETE.

C. DALCI script files

U031CSEC – update 03.1 Channel Security data source

The following DALCI script can be used to update all Channel Security records with appropriate default values for the new Data Encryption fields added as part of version 03.1. These default values can be modified for your specific requirements.

```
UPDATE CHANNEL_SECURITY
kkt_dt_hdr = ""
kkt_dt_keyx = ""
kkt_dt_kam = ""
kkt_dt_vvk = ""
kkt_dt_old_hdr = ""
kkt_dt_old_keyx = ""
kkt_dt_old_kam = ""
kkt_dt_old_vvk = ""
kkt_dt_variant = 0
kkt_dt_lgth = 1
kkt_old_dt_timr = ""
encrypt_des_mde = 0
encrypt_layers = 1
encrypt_data_typ = "U"
outbnd_iv_hdr = ""
outbnd_iv_keyx = ""
outbnd_iv_kam = ""
kam_lgth = 1
outbnd_kdt1_lv11_hdr = ""
outbnd_kdt1_lv11_keyx = ""
outbnd_kdt1_lv11_kam = ""
outbnd_kdt1_lv11_vvk = ""
outbnd_kdt1_lv12_hdr = ""
outbnd_kdt1_lv12_keyx = ""
```

```

outbnd_kdt1_lvl2_kam = ""
outbnd_kdt1_lvl2_vvk = ""
outbnd_kdt1_lvl1_old_hdr = ""
outbnd_kdt1_lvl1_old_keyx = ""
outbnd_kdt1_lvl1_old_kam = ""
outbnd_kdt1_lvl1_old_vvk = ""
outbnd_kdt1_lvl2_old_hdr = ""
outbnd_kdt1_lvl2_old_keyx = ""
outbnd_kdt1_lvl2_old_kam = ""
outbnd_kdt1_lvl2_old_vvk = ""
outbnd_kdt_lgth = 1
outbnd_old_data_timr =
0x2020202020202020202020202020202020202020202020202020202020202020 \
202020202020
;

```

You will be instructed when to perform the above in section 7.

U033CSEC – update 03.3 Channel Security data source

The following DALCI script can be used to update all Channel Security records with appropriate default values for the new Data Encryption fields added as part of version 03.3. This step should be run if you are a v02.4, v03.1, and v03.2 customer going to any release of TSS v03.3 and greater. These default values can be modified for your specific requirements.

```

UPDATE CHANNEL_SECURITY
kkt_am_export_vrnt = 0
;

```

You will be instructed when to perform the above in section 7.

U044CSEC – update 04.4 Channel Security data source

The following DALCI script can be used to update all Channel Security records with appropriate default values for the new fields added as part of version 04.4. These default values can be modified for your specific requirements.

```

UPDATE CHANNEL_SECURITY
outbnd_kam_lgth = "1"
chan_dev_typ = " "
;

```

You will be instructed when to perform the above in section 7.

U092CSEC – update 09.2 Channel Security data source

The following DALCI script can be used to update all Channel Security records with appropriate default values for the new fields added as part of BASE24-eps version 09.2. This step should be run if you are converting to version 09.2 or greater. These default values can be modified for your specific requirements.

```

UPDATE CHANNEL_SECURITY
inbnd_kam_key_typ = 0
outbnd_kam_key_typ = 0
inbnd_kdt_key_typ = 0
outbnd_kdt_key_typ = 0
inbnd_kdt1_lv11_hdr = ""
inbnd_kdt1_lv11_keyx = 0x00
inbnd_kdt1_lv11_kam = ""
inbnd_kdt1_lv11_vvk = ""
inbnd_kdt_lgth = "1"
inbnd_kdt1_lv11_old_hdr = ""
inbnd_kdt1_lv11_old_keyx = 0x00
inbnd_kdt1_lv11_old_kam = ""
inbnd_kdt1_lv11_old_vvk = ""
cross_acq_hdr = ""
cross_acq_keyx = 0x00
cross_acq_kam = ""
cross_acq_vvk = ""
cross_acq_lgth = "1"
acq_init_hdr = ""
acq_init_keyx = 0x00

```

```
acq_init_kam = ""
acq_init_vvk = ""
acq_init_lgth = "1"
init_kam_hdr = ""
init_kam_keyx = 0x00
init_kam_kam = ""
init_kam_vvk = ""
init_kam_lgth = "1"
mstr1_hdr = ""
mstr1_keyx = 0x00
mstr1_kam = ""
mstr1_vvk = ""
mstr1_lgth = "1"
mstr1_old_hdr = ""
mstr1_old_keyx = 0x00
mstr1_old_kam = ""
mstr1_old_vvk = ""
mstr2_hdr = ""
mstr2_keyx = 0x00
mstr2_kam = ""
mstr2_vvk = ""
mstr2_lgth = "1"
mstr2_old_hdr = ""
mstr2_old_keyx = 0x00
mstr2_old_kam = ""
mstr2_old_vvk = ""
inbnd_old_data_timr = ""
zka_drv_flg = 0
key_gen_num = ""
key_ver_num = ""
key_table_num = 1
drv_depth = 0
drv_id = ""
;
```

You will be instructed when to perform the above in section 7.

U111CSEC – update 11.1 Channel Security data source

The following DALCI script can be used to update all Channel Security records with appropriate default values for the new fields added as part of BASE24-eps version 11.1. This step should be run if you are converting to version 11.1 or greater. These default values can be modified for your specific requirements.

```
UPDATE CHANNEL_SECURITY
key_hierarchy = 2
kkt_opt = 1
mstr1_export_hdr = ""
mstr1_export_keyx = 0x00
mstr1_export_kam = ""
mstr1_export_vvk = ""
mstr_export_lgth = 1
inbnd_kdt1_lvl2_hdr = ""
inbnd_kdt1_lvl2_keyx = 0x00
inbnd_kdt1_lvl2_kam = ""
inbnd_kdt1_lvl2_vvk = ""
inbnd_kdt1_lvl2_old_hdr = ""
inbnd_kdt1_lvl2_old_keyx = 0x00
inbnd_kdt1_lvl2_old_kam = ""
inbnd_kdt1_lvl2_old_vvk = ""
futr_keys_usg = 0
;
```

You will be instructed when to perform the above in section 7.

U031ISEC – update 03.1 Interchange Security data source

The following DALCI script can be used to update all Interchange Security records with appropriate default values for the new Data Encryption fields added as part of version 03.1. These default values can be modified for your specific requirements.

```
UPDATE ICHG_SECURITY
kkt_dt_import_hdr = ""
kkt_dt_import_keyx = ""
kkt_dt_import_kam = ""
kkt_dt_import_vvk = ""
kkt_dt_export_hdr = ""
kkt_dt_export_keyx = ""
kkt_dt_export_kam = ""
kkt_dt_export_vvk = ""
kkt_dt_key_opt = 0x1
kkt_dt_lgth = 1
kkt_dt_variant = 0
inbnd_kdt1_lvl1_hdr = ""
inbnd_kdt1_lvl1_keyx = ""
inbnd_kdt1_lvl1_kam = ""
inbnd_kdt1_lvl1_vvk = ""
inbnd_kdt1_lvl2_hdr = ""
inbnd_kdt1_lvl2_keyx = ""
inbnd_kdt1_lvl2_kam = ""
inbnd_kdt1_lvl2_vvk = ""
inbnd_kdt2_lvl1_hdr = ""
inbnd_kdt2_lvl1_keyx = ""
inbnd_kdt2_lvl1_kam = ""
inbnd_kdt2_lvl1_vvk = ""
inbnd_kdt2_lvl2_hdr = ""
inbnd_kdt2_lvl2_keyx = ""
inbnd_kdt2_lvl2_kam = ""
inbnd_kdt2_lvl2_vvk = ""
inbnd_kdt_idx = 1
```


You will be instructed when to perform the above in section 7.

U032ISEC – update 03.2 Interchange Security data source

The following DALCI script can be used to update all Interchange Security records with appropriate default values for the new NCR Master Key fields added as part of version 03.2. These default values can be modified for your specific requirements.

```
UPDATE ICHG_SECURITY
kkt_pt_mstr_hdr = ""
kkt_pt_mstr_keyx = ""
kkt_pt_mstr_kam = ""
kkt_pt_mstr_vvk = ""
kkt_am_mstr_hdr = ""
kkt_am_mstr_keyx = ""
kkt_am_mstr_kam = ""
kkt_am_mstr_vvk = ""
;
```

U092ISEC – update 09.2 Interchange Security data source

The following DALCI script can be used to update all Interchange Security records with appropriate default values for the new BASE24-eps fields added as part of version 09.2. These default values can be modified for your specific requirements.

Since these fields were renamed between releases 09.2 and 11.1, your upgrade will require only one of the two following DALCI scripts depending on the release to which you are upgrading.

If you are upgrading to 09.2, use the following DALCI scripts for U092ISEC

```
UPDATE ICHG_SECURITY
inbnd_kam_key_typ = 0
outbnd_kam_key_typ = 0
inbnd_kdt_key_typ = 0
outbnd_kdt_key_typ = 0
```

```
import_kkt_key_typ = 0
export_kkt_key_typ = 0
kkt_pend_pt_import_hdr = ""
kkt_pend_pt_import_keyx = 0x00
kkt_pend_pt_import_kam = ""
kkt_pend_pt_import_vvk = ""
kkt_pend_am_import_hdr = ""
kkt_pend_am_import_keyx = 0x00
kkt_pend_am_import_kam = ""
kkt_pend_am_import_vvk = ""
kkt_pend_dt_import_hdr = ""
kkt_pend_dt_import_keyx = 0x00
kkt_pend_dt_import_kam = ""
kkt_pend_dt_import_vvk = ""
kkt_pend_pt_export_hdr = ""
kkt_pend_pt_export_keyx = 0x00
kkt_pend_pt_export_kam = ""
kkt_pend_pt_export_vvk = ""
kkt_pend_am_export_hdr = ""
kkt_pend_am_export_keyx = 0x00
kkt_pend_am_export_kam = ""
kkt_pend_am_export_vvk = ""
kkt_pend_dt_export_hdr = ""
kkt_pend_dt_export_keyx = 0x00
kkt_pend_dt_export_kam = ""
kkt_pend_dt_export_vvk = ""
input_key_typ = 0
zka_drv_flg = 0
key_gen_num = ""
key_ver_num = ""
key_table_num = 1
drv_depth = 0
drv_id = ""
;
```

If you are upgrading to 11.1, use the following DALCI scripts for U092ISEC

```
UPDATE ICHG_SECURITY
inbnd_kam_key_typ = 0
outbnd_kam_key_typ = 0
inbnd_kdt_key_typ = 0
outbnd_kdt_key_typ = 0
import_kkt_key_typ = 0
export_kkt_key_typ = 0
kkt_pt_import_old_hdr = ""
kkt_pt_import_old_keyx = 0x00
kkt_pt_import_old_kam = ""
kkt_pt_import_old_vvk = ""
kkt_am_import_old_hdr = ""
kkt_am_import_old_keyx = 0x00
kkt_am_import_old_kam = ""
kkt_am_import_old_vvk = ""
kkt_dt_import_old_hdr = ""
kkt_dt_import_old_keyx = 0x00
kkt_dt_import_old_kam = ""
kkt_dt_import_old_vvk = ""
kkt_pt_export_old_hdr = ""
kkt_pt_export_old_keyx = 0x00
kkt_pt_export_old_kam = ""
kkt_pt_export_old_vvk = ""
kkt_am_export_old_hdr = ""
kkt_am_export_old_keyx = 0x00
kkt_am_export_old_kam = ""
kkt_am_export_old_vvk = ""
kkt_dt_export_old_hdr = ""
kkt_dt_export_old_keyx = 0x00
kkt_dt_export_old_kam = ""
kkt_dt_export_old_vvk = ""
input_key_typ = 0
zka_drv_flg = 0
key_gen_num = ""
```

```
key_ver_num = ""
key_table_num = 1
drv_depth = 0
drv_id = ""
;
```

You will be instructed when to perform the above in section 7.

U111ISEC – update 11.1 Interchange Security data source

The following DALCI script can be used to update all Interchange Security records with appropriate default values for the new BASE24-eps fields added as part of version 11.1. These default values can be modified for your specific requirements.

```
UPDATE ICHG_SECURITY
key_hierarchy = 2
kkt_opt = 1
mstr1_export_hdr = ""
mstr1_export_keyx = 0x00
mstr1_export_kam = ""
mstr1_export_vvk = ""
mstr1_import_hdr = ""
mstr1_import_keyx = 0x00
mstr1_import_kam = ""
mstr1_import_vvk = ""
mstr1_kkt_lgth = 1
mstr1_import_kkt_pt_hdr = ""
mstr1_import_kkt_pt_keyx = 0x00
mstr1_import_kkt_pt_kam = ""
mstr1_import_kkt_pt_vvk = ""
mstr1_kpt_lgth = 1
mstr1_import_kkt_am_hdr = ""
mstr1_import_kkt_am_keyx = 0x00
mstr1_import_kkt_am_kam = ""
mstr1_import_kkt_am_vvk = ""
```

```
mstr1_kam_lgth = 1
mstr1_import_kkt_dt_hdr = ""
mstr1_import_kkt_dt_keyx = 0x00
mstr1_import_kkt_dt_kam = ""
mstr1_import_kkt_dt_vvk = ""
mstr1_kdt_lgth = 1
old_kek_timr = "0x20"
futr_keys_usg = 0
multi_flg = N
series_enabled_flg = N
;
```

You will be instructed when to perform the above in section 7.



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